PhD. Proposal Defense

Dynamic Bayesian Networks for Data-Driven Fault Diagnosis and Prognosis of Building Systems

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    Objective 2: Development of Fault Diagnosis Strategy for Grid-Interactive Efficient Buildings (GEBs)

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Introduction

48% energy consumption in US by building sector

US EIA Global Status Report 2018

1.5% estimated annual increase for next 20 years

US EIA Global Status Report 2018

30% wasted energy due to malfunctioning controls and equipment; operation faults

Deshmukh 2018; Fernandez 2017; Granderson 2017a; Katipamula 2005; Roth 2005; Wall 2018

40% potential energy saving for HVAC systems with fault detection and diagnosis (FDD)

Deshmukh 2018; Fernandez 2017; Granderson 2017a; Katipamula 2005; Roth 2005; Wall 2018

Research Needs

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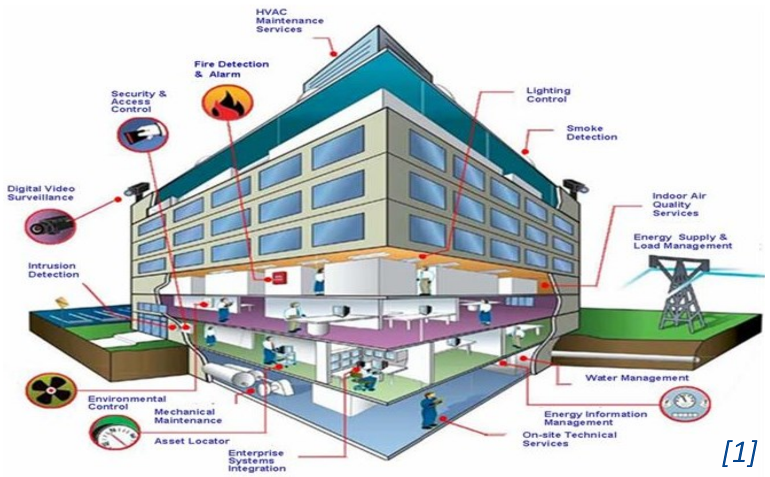
Building Automation System (BAS) Internet of Things (IoT) Sensors, meters etc.	Surge in Data Analytics and Machine Learning Tools	Increased Interactivity with Power Grid



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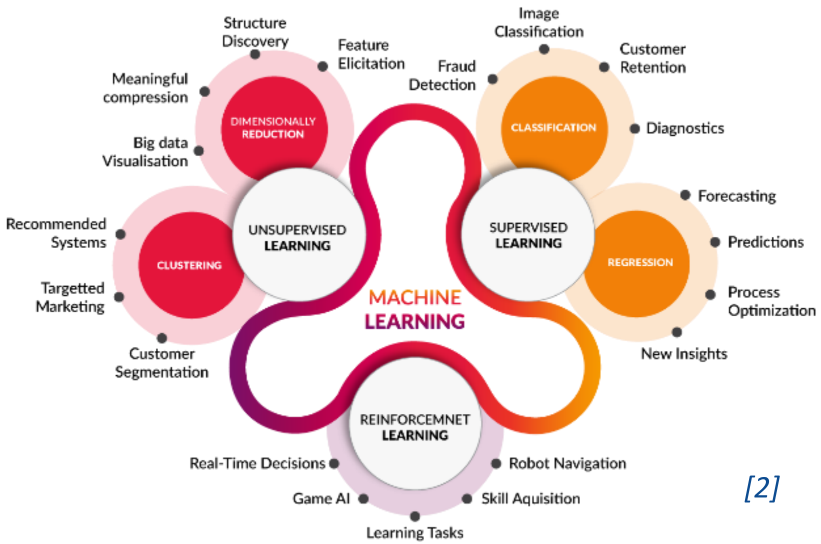

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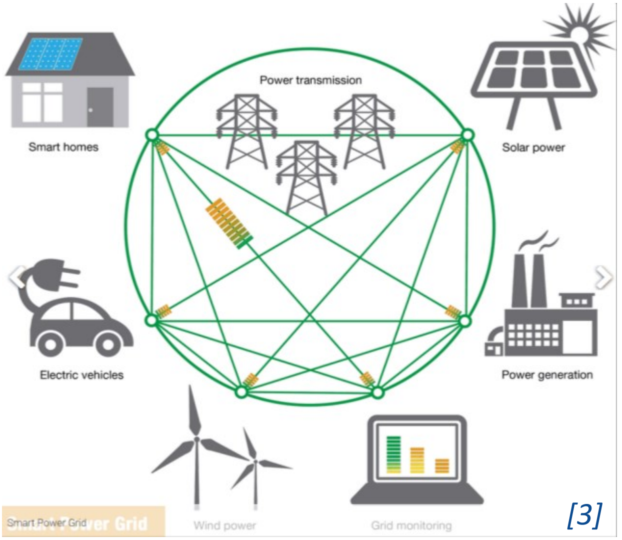

[3]

- ↑ Large amount of data collected
- ↑ High level of automation
- ↑ Deployment of continuous commissioning tools

- ↑ Simple, cost-effective solutions
- ↑ Low engineering effort
- ↑ Learn intricate relationships
- ↓ Largely dependent on data quality

- ↑ Provides demand flexibility
- ↑ Resilience and comfort
- ↓ Vulnerable to active threats

[1] http://www.promptautomation.com/building-automation
[2] https://www.cognub.com/index.php/cognitive-platform
[3] http://www.news.gatech.edu/features/building-power-grid-future

Research Gaps

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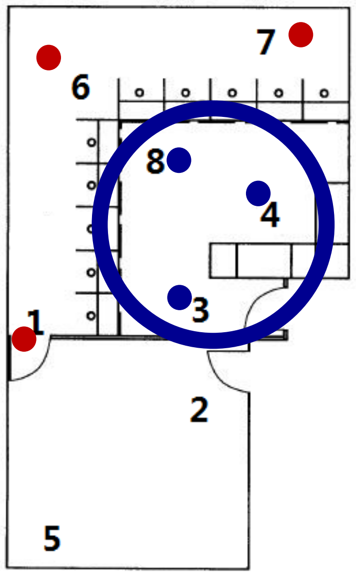
- Poor data quality <u>impedes</u> the development and application of data-driven methods

Databases obtained from BAS may be incomplete due to:

- Equipment failures
- Communication/transmission issues
- Data corruption
- Inconsistencies due to sensor noise

Loss of valuable information

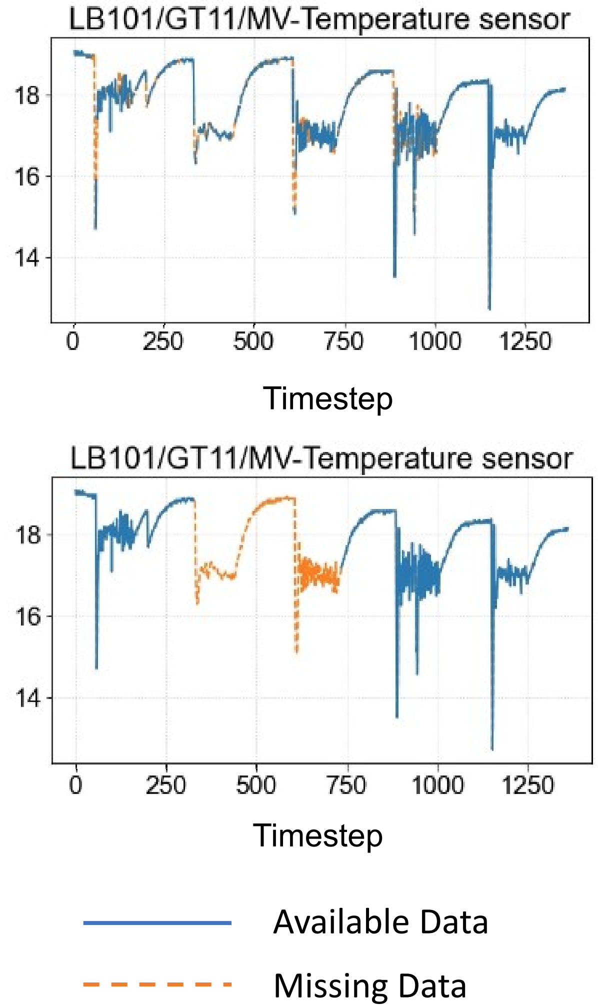


Liu et al., 2018

Building measurements are often <u>strongly correlated</u>

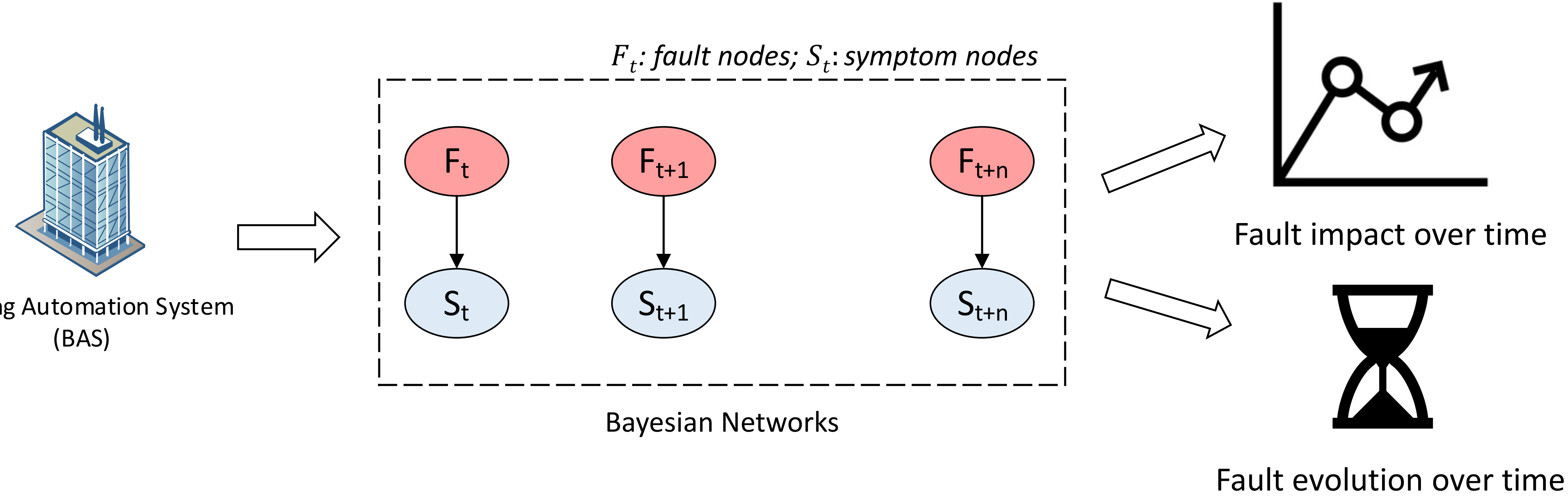
Data Imputation

Gap 1: Multivariate data imputation methods to defy poor data quality have not been utilized efficiently



Research Gaps

- Representing temporal dependencies in diagnostic models is challenging



Labeled ground truth data that confirms the root fault causes are hard to obtain

Gap 2: Lack of a systematic and scalable fault diagnosis strategy to track fault evolution

Research Gaps

- Integration of building technologies and network-based communication systems have made buildings vulnerable to active threats such as cyber-attacks

[1]

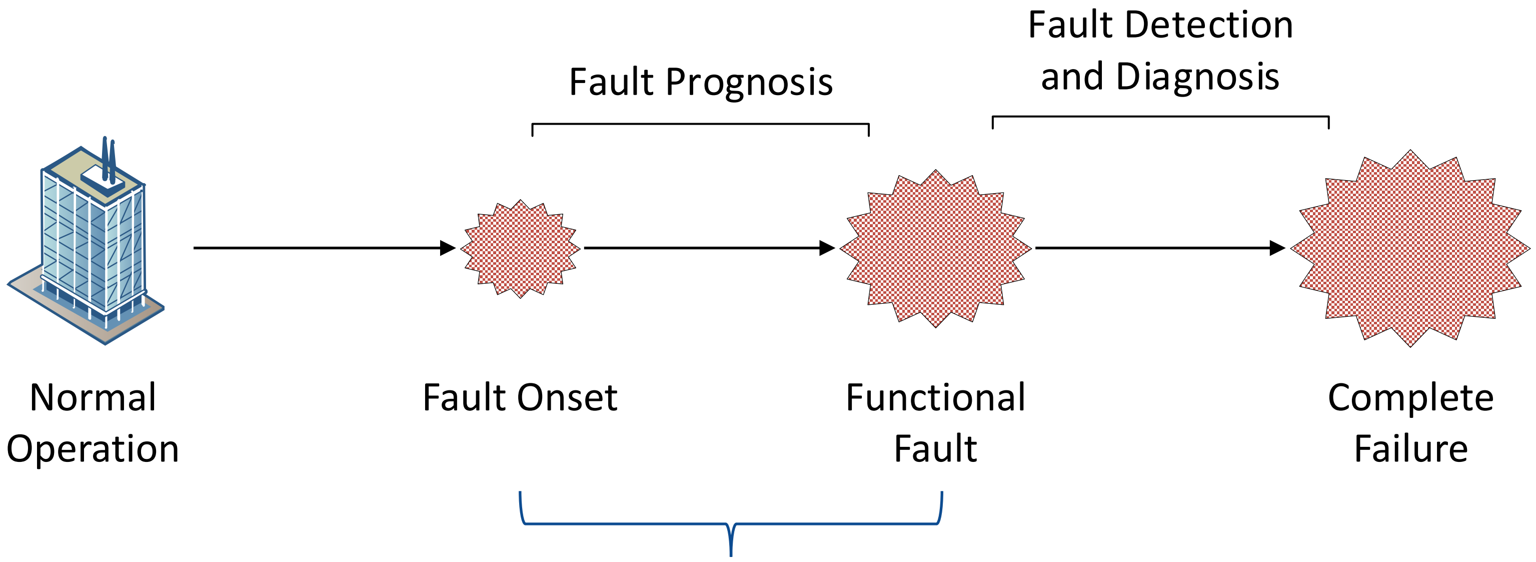
- Building Automation and Control networks (BACnet) protocol is not designed with security requirements *(Fu et al., 2021)*
- Vulnerability in one component can be used to compromise other components
- Recent attacks:
  - Google's Sydney headquarter in 2013 [2]
  - Target system in 2014 [3]
  - Hotel in Austrian Alps in 2017 [4]

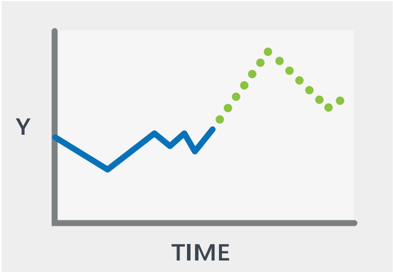
Gap 3: Active threats information are typically not incorporated in a FDD framework

[1] https://ingenuity.siemens.com/2020/07/the-top-five-trends-shaping-cybersecurity-in-smart-buildings
[2] https://www.nbcnews.com/id/wbna51805522
[3] https://www.computerworld.com/article/2487452/target-attack- shows-danger-of-remotely-accessible-hvac-systems
[4] https://www.bbc.com/news/business-42352326

Research Gaps

- Fault detection and diagnosis (FDD) is <u>insufficient</u> in cases where critical functions of the system may have already failed



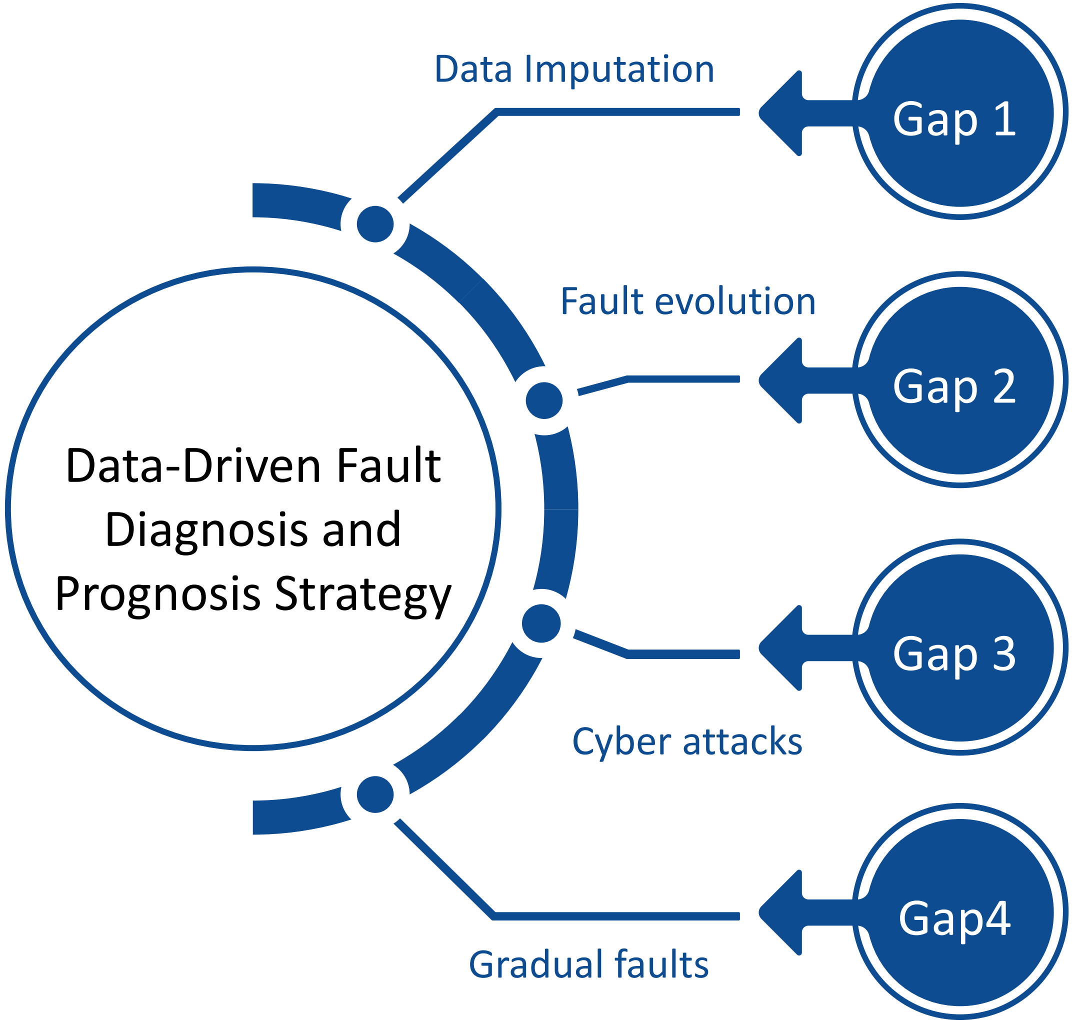


Predict future faults probabilities

Estimated that over 20% of HVAC systems are running under early stage of gradual faults *(Yu et al., 2014)*

Gap 4: Fault prognosis strategies to preemptively identify gradual faults are hardly studied

Research Objectives



Objective 1: Evaluate reported data imputation methods and their impact on data-driven algorithms

Objective 2: Develop a systematic and scalable Dynamic Bayesian Network (DBN) framework for cyber-physical fault diagnosis in GEBs

Objective 3: Develop a Dynamic Bayesian Network (DBN) framework for fault prognosis

Objective 1: Data Quality and Data Imputation

Task 1.1 Survey potential data imputation methods that utilize mutual information to handle missing BAS data

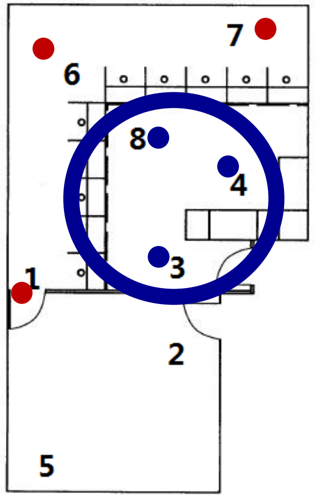
Task 1.2 Evaluate data imputation methods suitable for multi-source BAS data

Task 1.3 Evaluate the impact of the repaired dataset on the performance of data-driven algorithms

Task 1.1 Multivariate Data Imputation

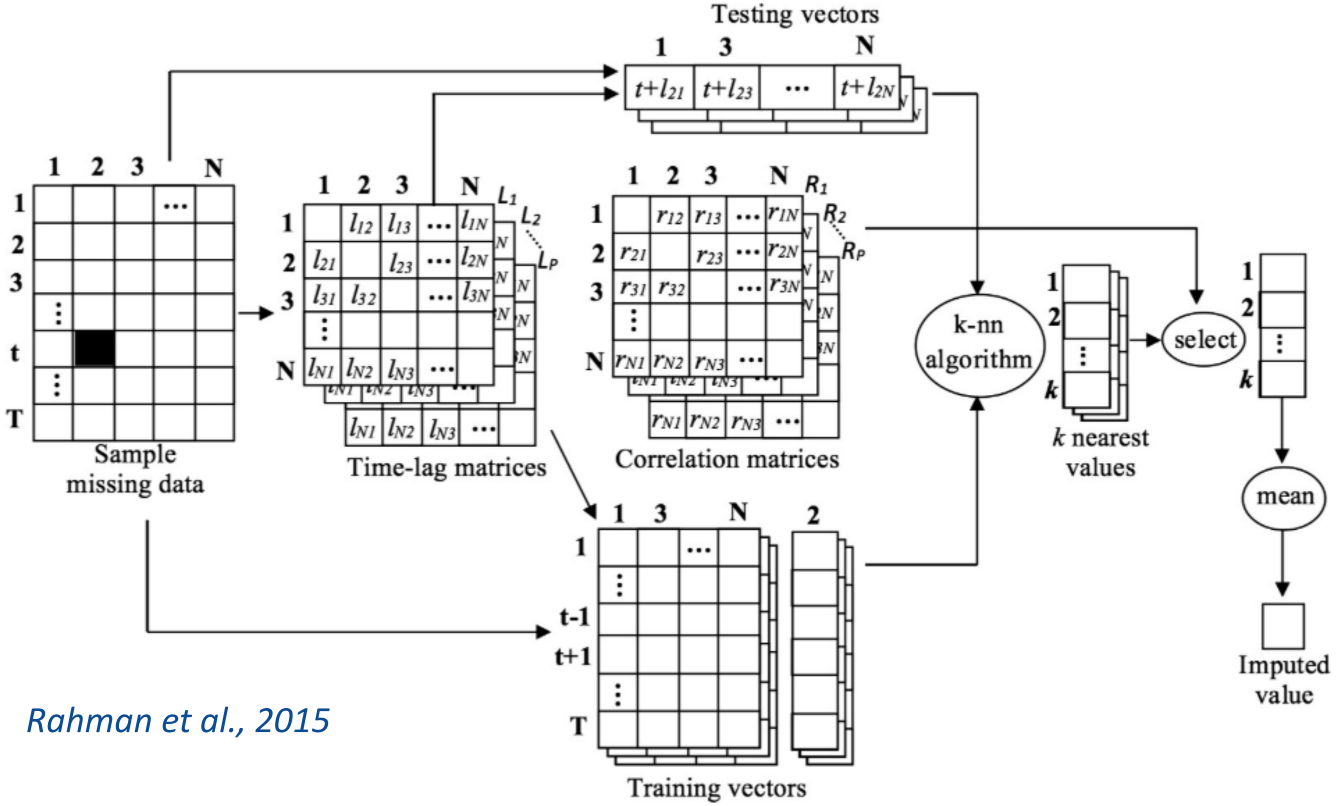
Challenges: 1. Leveraging mutual information within a multivariate dataset

2. Identifying correlations across time



Liu et al., 2018

Lagged k-Nearest Neighbor (kNN) Approach:



Rahman et al., 2015

1. Calculate time-lags (p = 6)
2. Calculate cross-correlations
3. Form training and testing vectors
4. Find nearest neighbors (k = 5)
5. Impute missing values

Task 1.2 & 1.3 Evaluation

FHS Building (Stockholm, Sweden)

- Mixed-use institution building
- Multiple data sources
- District heating, cooling and electric supply

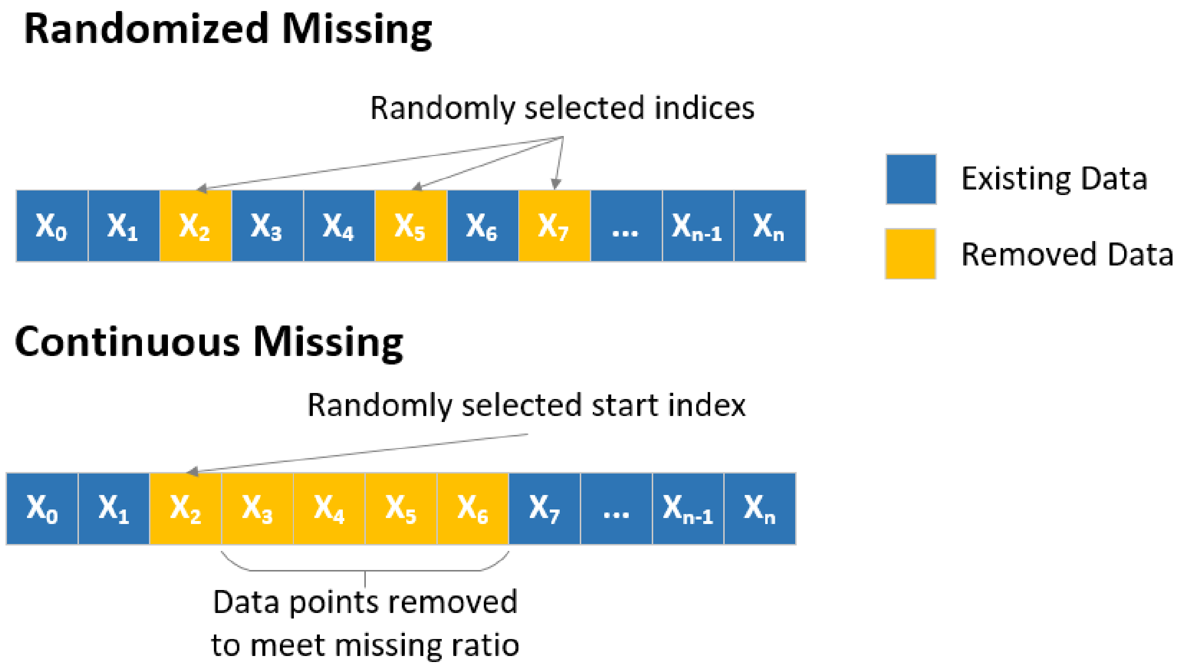
Experimental Data

- Selected 13 sensors from FHS building
- Operation data collected from March 1-7, 2021 (5-min sampling interval)

No.	System	Description
1	BAS	AHU supply air static pressure
2	BAS	AHU supply air flowrate
3	BAS	AHU supply air temperature
4	BAS	AHU supply fan speed
5	BAS	AHU cooling coil valve position
6	BAS	Radiator hot water temperature
7	Weather Station	Outdoor humidity
8	Weather Station	Outdoor temperature
9	DCV	Actual supply-air flow
10	DCV	Actual supply-air temperature
11	DCV	Actual supply-air pressure
12	DCV	Actual room temperature
13	DCV	Actual carbon dioxide concentration

Task 1.2 & 1.3 Evaluation

Validation Dataset (10-40% missing ratio)



Methods Compared:

1. Linear Interpolation
2. kNN Imputer
3. Lagged-kNN
4. Fourier Imputation
5. Combined Fourier Lagged-kNN (FL-kNN)

Evaluation Criteria:

1. Imputation Accuracy
2. Impact on Energy Forecasting

Performance Metric:

Normalized Root Mean Squared Error (NRMSE)

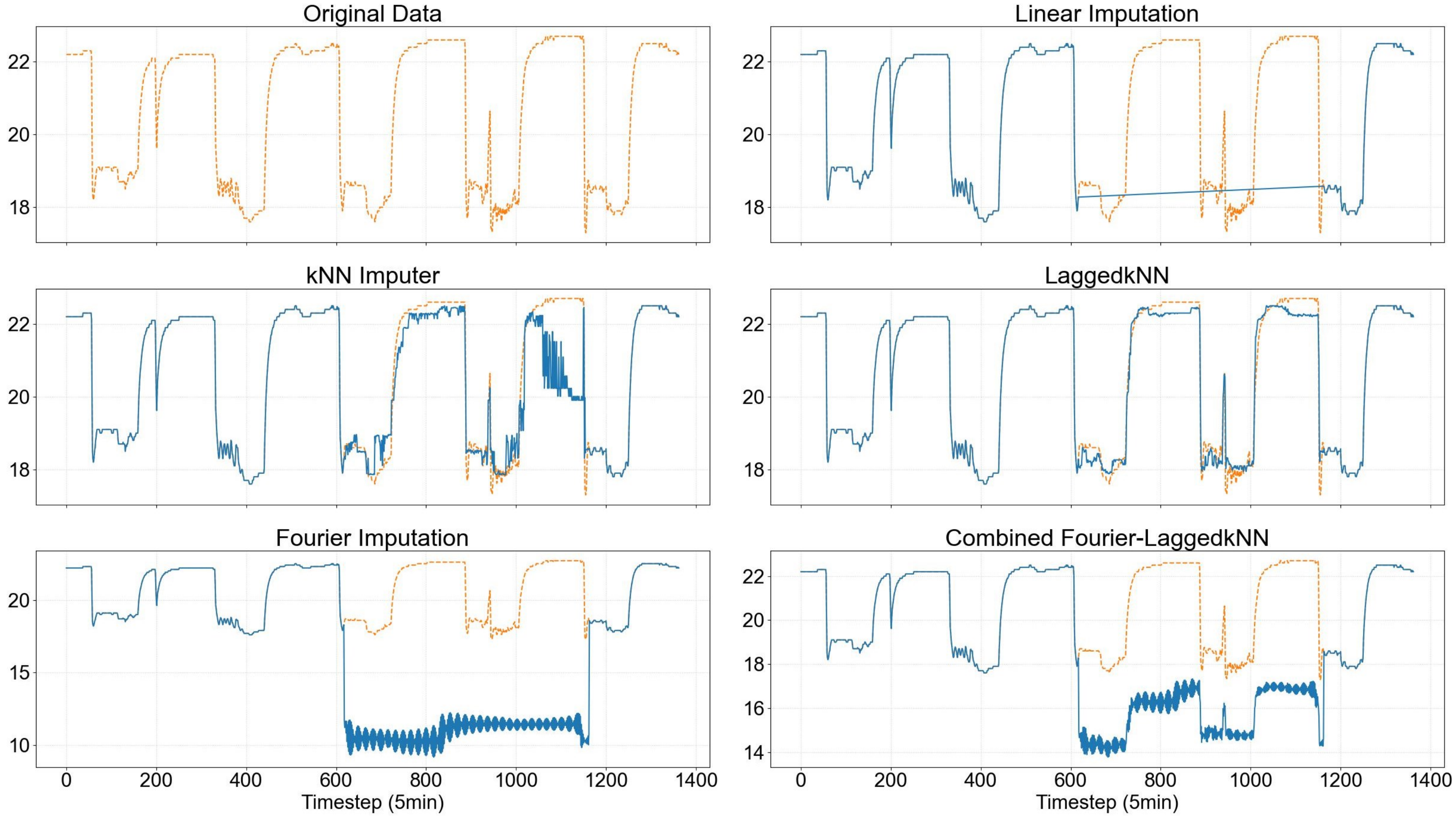
Task 1.2 & 1.3 Evaluation

Validation Dataset	Averaged Imputation Error					Averaged Energy Forecasting Error				
	Linear	kNN Imputer	Lagged kNN	Fourier	Fourier-Lagged kNN	Linear	kNN Imputer	Lagged kNN	Fourier	Fourier-Lagged kNN
10%-Continuous	0.772	0.291	**0.207**	1.229	0.664	0.025	0.025	**0.024**	0.025	0.025
20%-Continuous	0.630	0.486	**0.273**	0.935	0.525	0.042	0.038	**0.036**	0.038	0.039
30%-Continuous	0.874	0.436	**0.338**	1.092	0.594	0.066	0.076	**0.063**	0.074	0.072
40%-Continuous	1.089	0.455	**0.210**	1.113	0.545	0.082	0.091	**0.072**	0.076	0.079
10%-Random	0.087	0.161	0.098	0.111	**0.081**	**0.021**	**0.021**	**0.021**	0.022	**0.021**
20%-Random	0.105	0.193	**0.063**	0.166	0.099	**0.020**	**0.020**	**0.020**	**0.020**	**0.020**
30%-Random	0.127	0.292	**0.086**	0.257	0.153	0.024	0.024	0.024	0.024	**0.023**
40%-Random	0.118	0.430	**0.093**	0.269	0.160	0.025	0.025	**0.024**	0.025	0.025

- Lagged-kNN method has best performance in 7 out of 8 validation sets
- For the combined method, inaccuracies from Fourier imputation leads to larger overall error
- Statistical methods do not perform well for high missing ratios

Task 1.2 & 1.3 Evaluation

Actual Supply-Air Temperature (40% Continuously Missing)



Conclusions of Objective 1

- Potential in utilizing information from correlated variables to impute missing values in BAS data
- kNN methods have potential to cluster correlated variables
- Incorporating time-lagged cross correlations help improve imputation accuracy

Future Tasks:

- Assess the nearest neighbors (strongly correlated variates) identified from the lagged-kNN algorithm against domain knowledge
- Use information entropy to evaluate the quality of repaired dataset
- Evaluate the impact of artificially repaired data on fault detection methods

Objective 2: Development of Fault Diagnosis Strategy for GEBs

Task 2.1 Develop DBN-based diagnosis framework with discretized temporal conditional probability parameters

Task 2.2 Develop and integrate an information entropy-based historical baseline establishment approach

Task 2.3 Evaluate the performance and scalability of the framework using data collected from a real building, laboratory building and virtual testbed

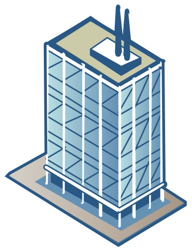
Task 2.1 Fault Diagnosis

Challenge: accurate reasoning of the HVAC system and its control strategies is required

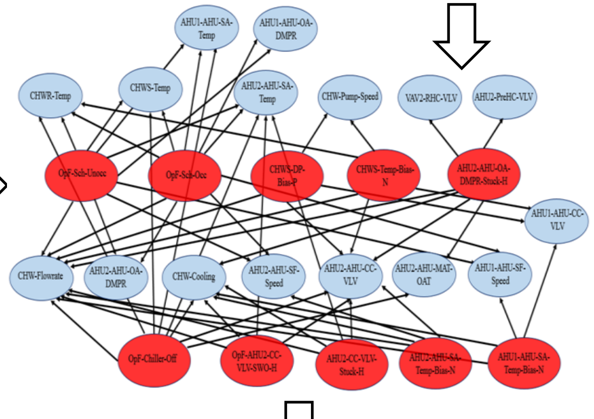
Bayesian Networks (BN): estimate fault probabilities based on observations from BAS measurements

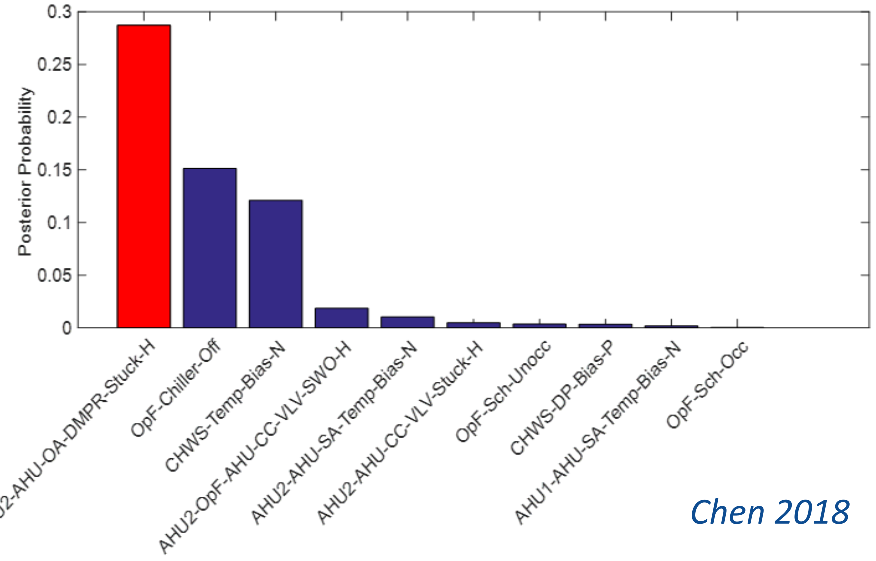
- Najafi et al., introduced BN for fault diagnosis in AHUs *(2012)*
- Zhao et al., proposed component-level diagnostic BN for 28 different faults in AHUs *(2015; 2017)*
- Xiao et al., developed a diagnostic BN for typical VAV terminal *(2014)*
- Chen et al., developed diagnostic BN strategy for whole-building datasets with high data dimensionality *(2018)*

Domain Knowledge



Building Automation System (BAS)





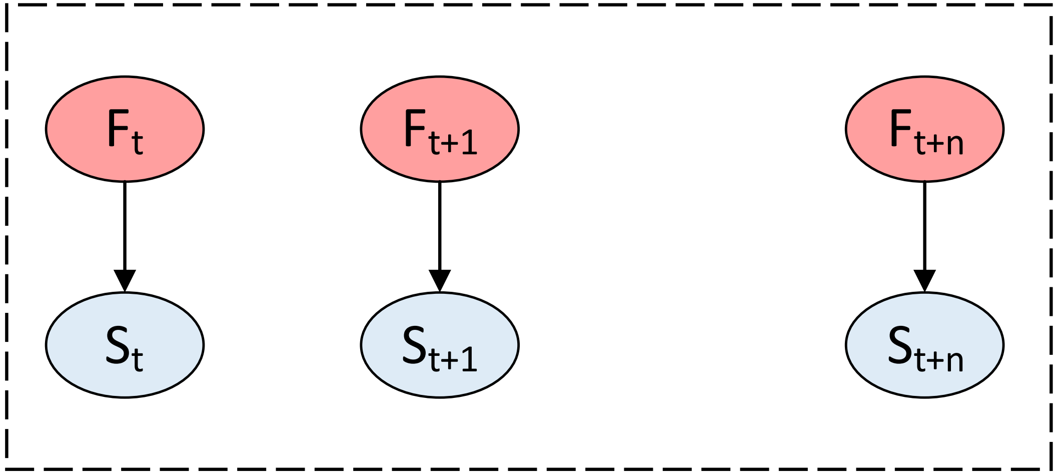
Chen 2018

Task 2.1 DBN Overview

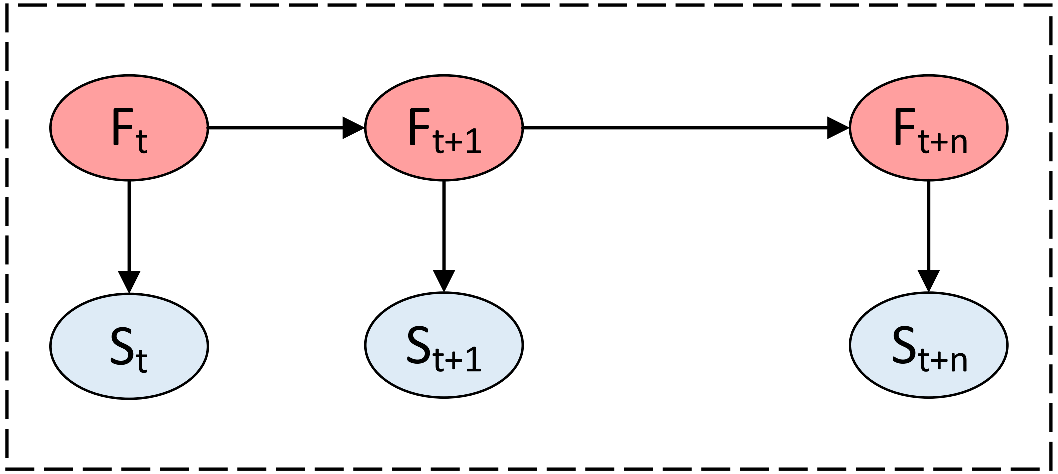
Dynamic Bayesian Networks (DBN):

- Extension of BN; more suitable for continuous system *(Murphy 2002)*
- DBN carries over past information allowing fault beliefs to accumulate over time
- Reduced false positive rate and enhanced fault belief when using a DBN *(Shi et al., 2018)*

F_t: fault nodes; S_t: symptom nodes



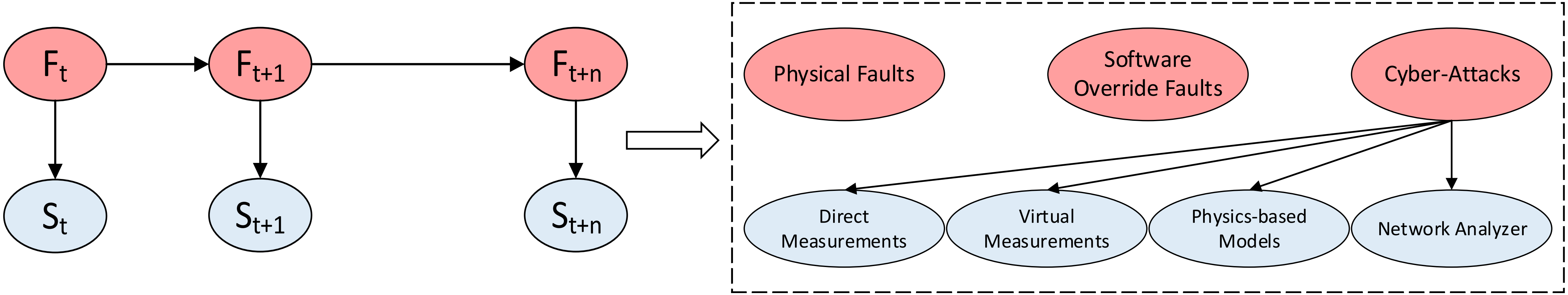
Static BN: $F(t+1)$ has no dependency on $F(t)$ from previous timestep



Dynamic BN: $F(t+1)$ has dependency on $F(t)$ from previous timestep

Task 2.1 DBN Structure Model

Developed using domain knowledge, physical analysis and physics-based models



Direct Measurements: $T_{SA},\ SPD_{SA}, Flow_{RA}$

Virtual Measurements: $(T_{SA}-T_{SA,\,STP}), OA_{fraction}$

T: temperature; SPD: speed; STP: setpoint
SA: supply air; OA: outdoor air; RA: return air

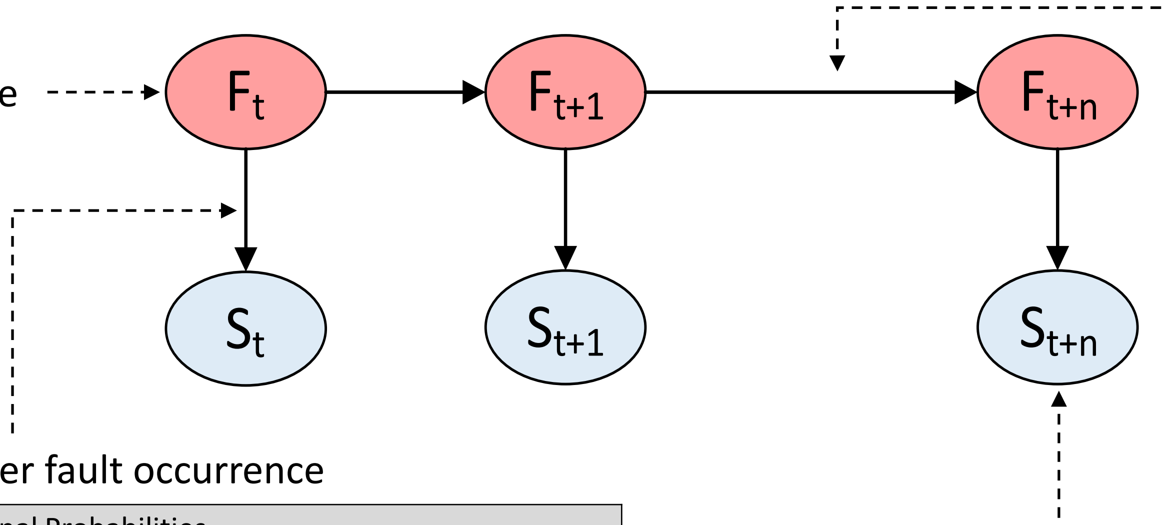
Physics-based models: Return fan curves (efficiency vs. flow)

Network Analyzer: Network attack detection probability

In total, 38 fault nodes and 28 symptom (evidence) nodes are included in the DBN

Task 2.1 DBN Parameter Model

Literature supported parameter model based on domain knowledge *(Regnier & Wen 2016; Chen 2018)*



1) Frequency of fault occurrence

Prior Probabilities	
Faulty	0.01
Fault Free	0.99

2) Probability of symptoms under fault occurrence

Conditional Probabilities		
Relationship Strength	Severity Level	Conditional Probability Under Fault
Strong Evidence	Very Severe	0.45
	Severe	0.45
	Low Severity	0.1
Moderate Evidence	Very Severe	0.25
	Severe	0.25
	Low Severity	0.5
Weak Evidence	Very Severe	0.05
	Severe	0.05
	Low Severity	0.9

3) Probability of fault occurrence into the next timestep

Temporal Conditional Probabilities		
Fault Node (t-1)	Faulty	Fault Free
Faulty	0.99	0.01
Fault Free	0.01	0.99

4) Probability of symptoms under no fault occurrence

$$LEAK\ Probability = \frac{Number\ of\ outlier\ data\ sample}{Total\ number\ of\ data\ sample}$$

Task 2.2 Historical Baseline Establishment

Since building's operation varies based on outdoor weather and internal load conditions:

- Necessary to artificially construct a baseline using historical data
- Reported methods are <u>not fully data-driven </u>and often <u>require expert knowledge</u>

Cosine Correlation Information Entropy (CCIE) Approach

Phase I: Match Weather Conditions	Phase II: Determine Baseline Samples

(Huang et al., 2022)

Historical Fault-Free Data

Incoming Real-time Data

Match Enthalpy

Determine, rank and select samples by enthalpy difference

Baseline Candidate Pool

Select samples to maximize correlation information entropy

Final Baseline

- Automatically constructs and evaluates the baseline
- Successfully implemented for fault detection
- Requires less samples than other reported methods

Task 2.3 Evaluation: Summary

Nesbitt Hall (Drexel University)

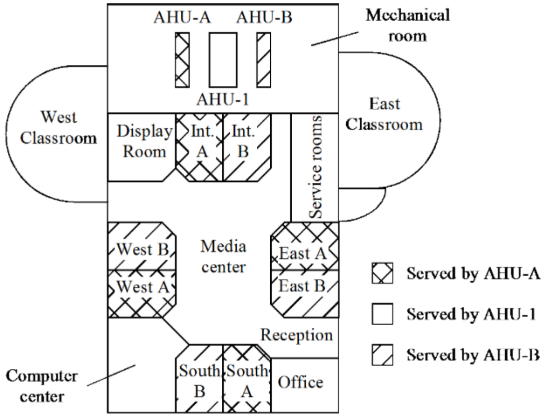
- Mixed-use institution building
- Water-cooled chillers
- Multi-zone VAV
    - Normal (fault-free) operation data
    - 14 artificially injected fault cases

Chen 2018; Zhang 2018

Laboratory Test Facility (ASHRAE RP-1312)

- Small commercial building
- Twin AHU system (AHU-A and B)
    - AHU-B running in fault-free operation
    - 11 artificially injected fault cases

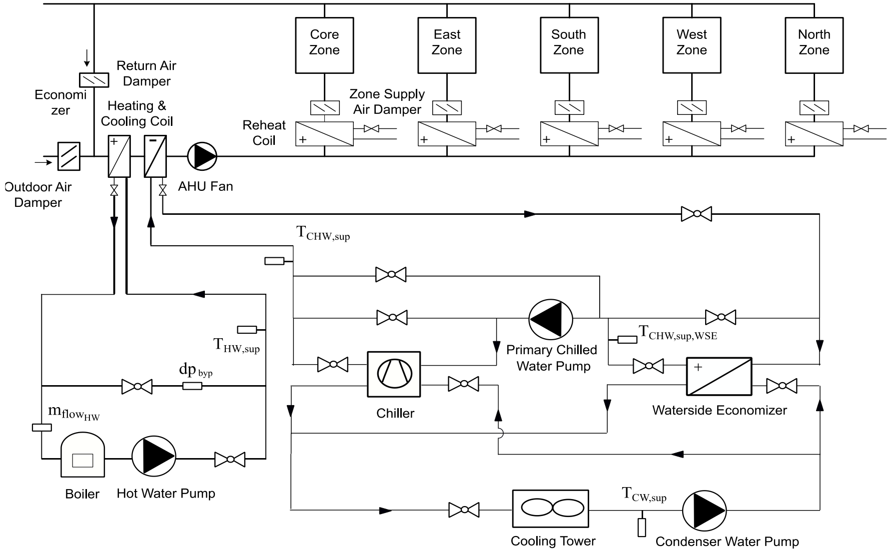
ERS Test Facility
Wen & Li, 2011



Virtual Testbed (Medium Office)

- One AHU connected with 5 VAV terminals serving 5 zones
- ASHRAE's Guideline 36 control sequence *(ASHRAE 2018)*
- Integrated with Hardware-in-the-Loop (HIL) testbed
- Both physical faults and cyber-attacks

Fu et al., 2021; Lu et al., 2021



Task 2.3 Evaluation: Summary

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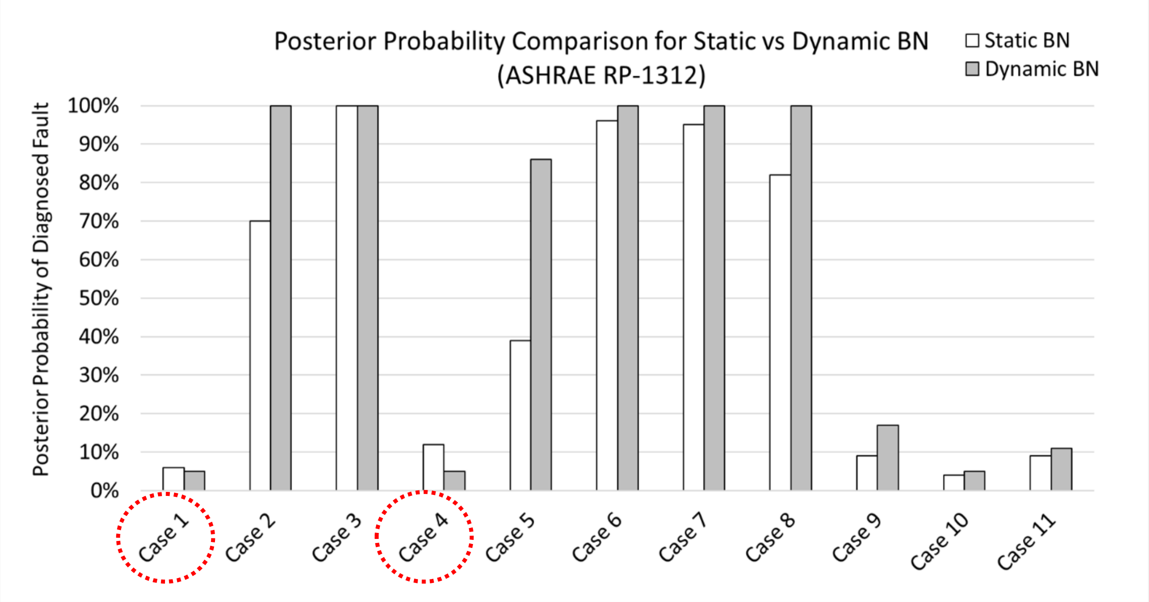
Framework Performance and Scalability

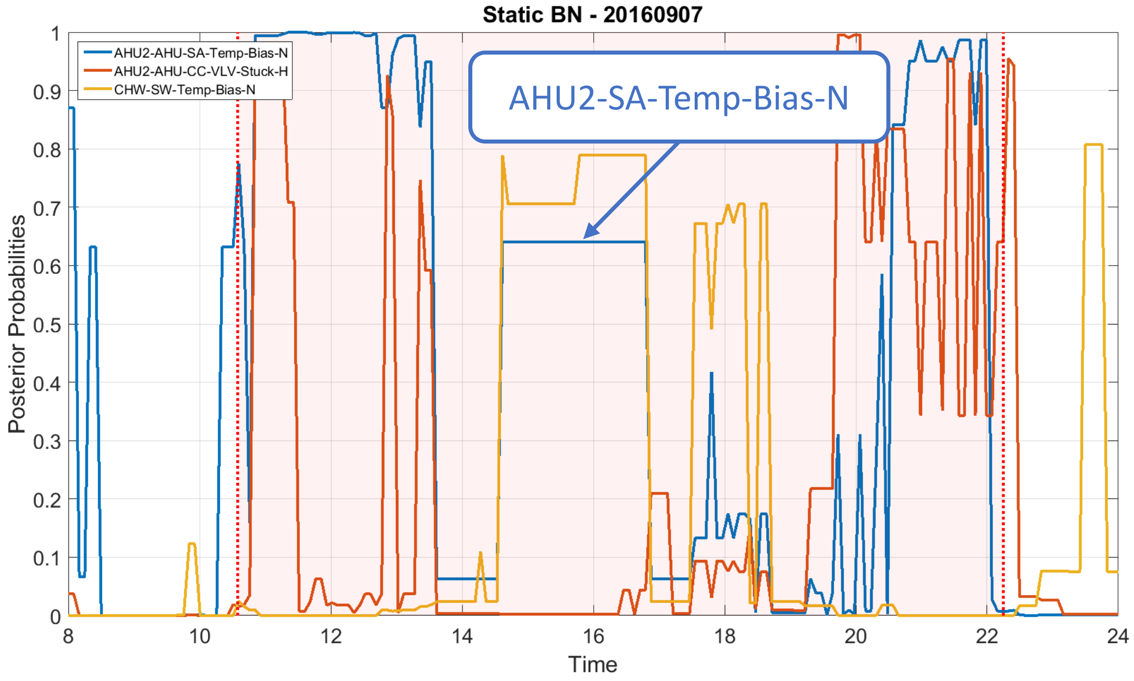
Dataset	Fault Configurations	Control Sequence	Total Cases	Diagnosed	Misdiagnosed Cases
Nesbitt	Physical Faults Injected in Real Building	Traditional	14	11	No fault symptom (1); Simultaneous faults (2)
ASHRAE RP-1312	Physical Faults Injected in Laboratory		11	11	-
MedOffice-Phy	Simulated Physical Faults	ASHRAE Guideline 36	13	13	-
MedOffice-Cyb	Simulated Cyber Attacks causing Physical Faults		4	3	No fault symptom (1)
MedOffice-PhyHIL	Physical Faults Injected in HIL Testbed		2	2	-
MedOffice-CybHIL	Cyber Attacks Injected in HIL Testbed		3	3	-

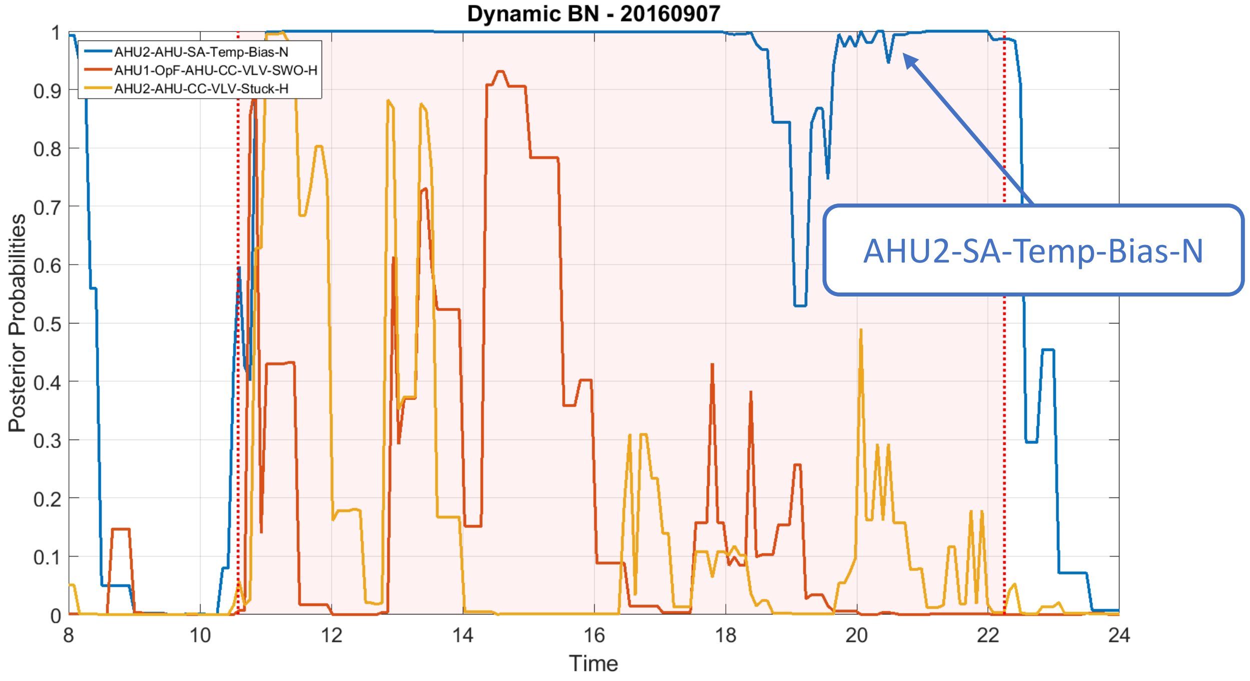


- Increased fault belief for diagnosed fault with DBN
- Reduced false positives for fault-free cases

Task 2.3 Evaluation: Real Building

AHU-2 SA Temp Negative Bias fault (Top 3 ranked faults)



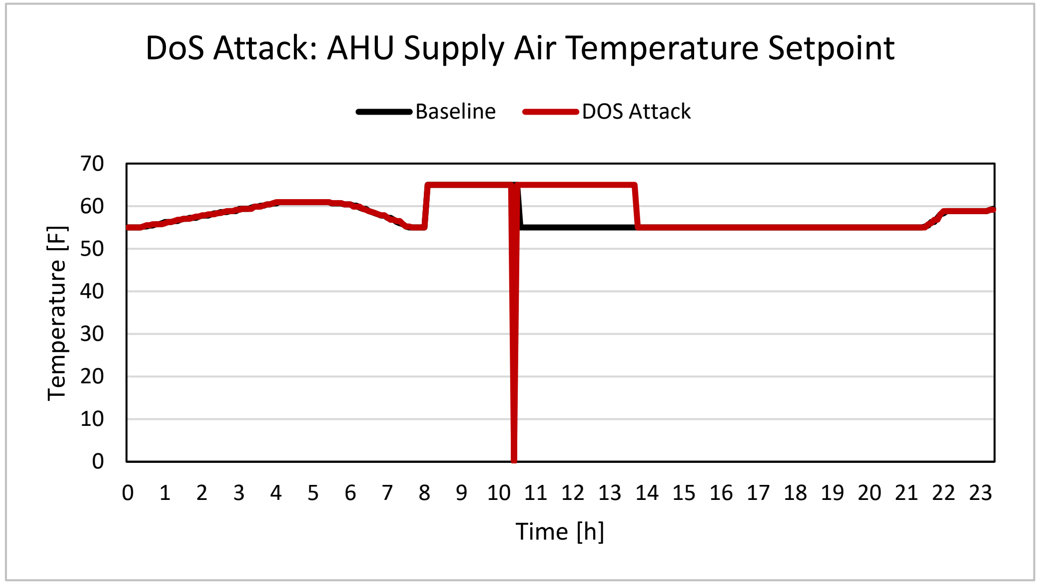


- Stronger fault belief observed with DBN inside the fault window

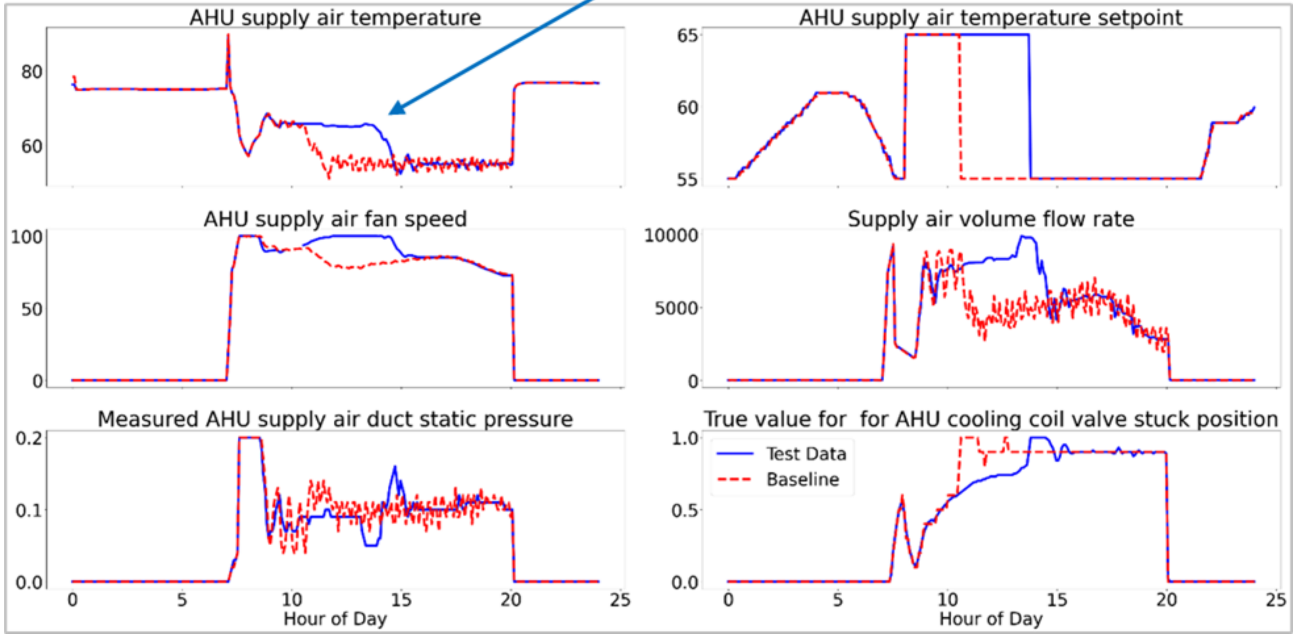
- DBN is more sensitive to discrepancies between incoming and baseline data

Task 2.3 Evaluation: HIL Testbed

Cyber Attack - Network DoS attack on AHU



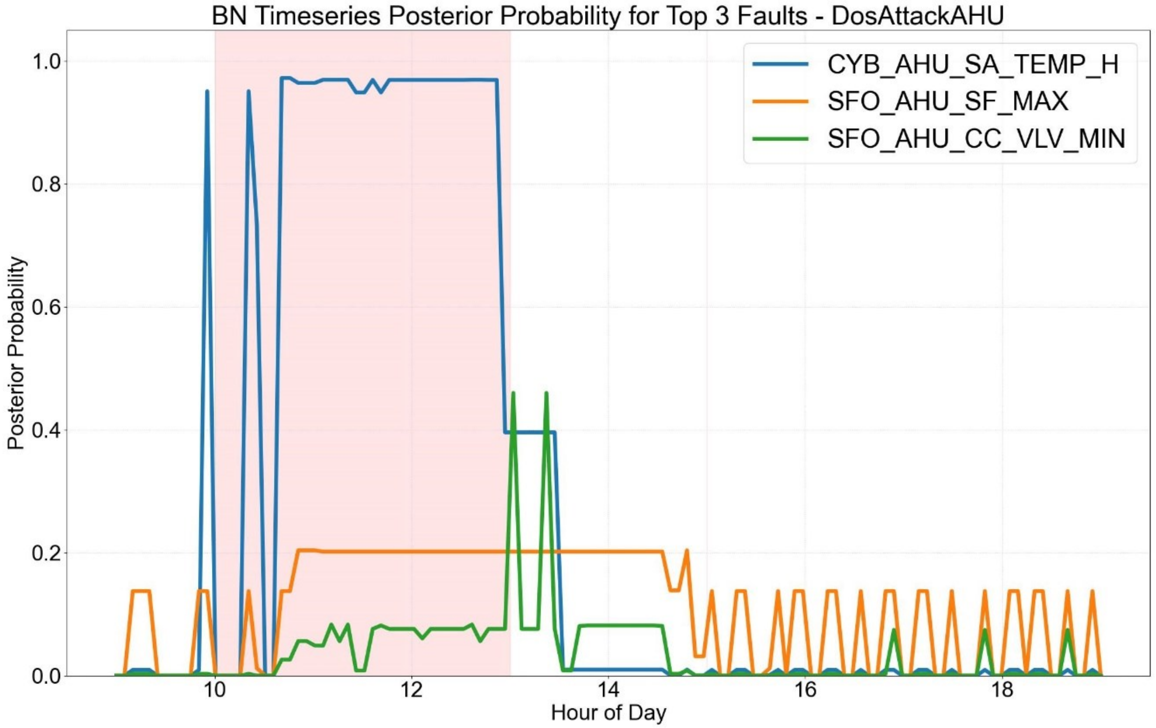
AHU supply air temperature abnormality continues until ~3pm

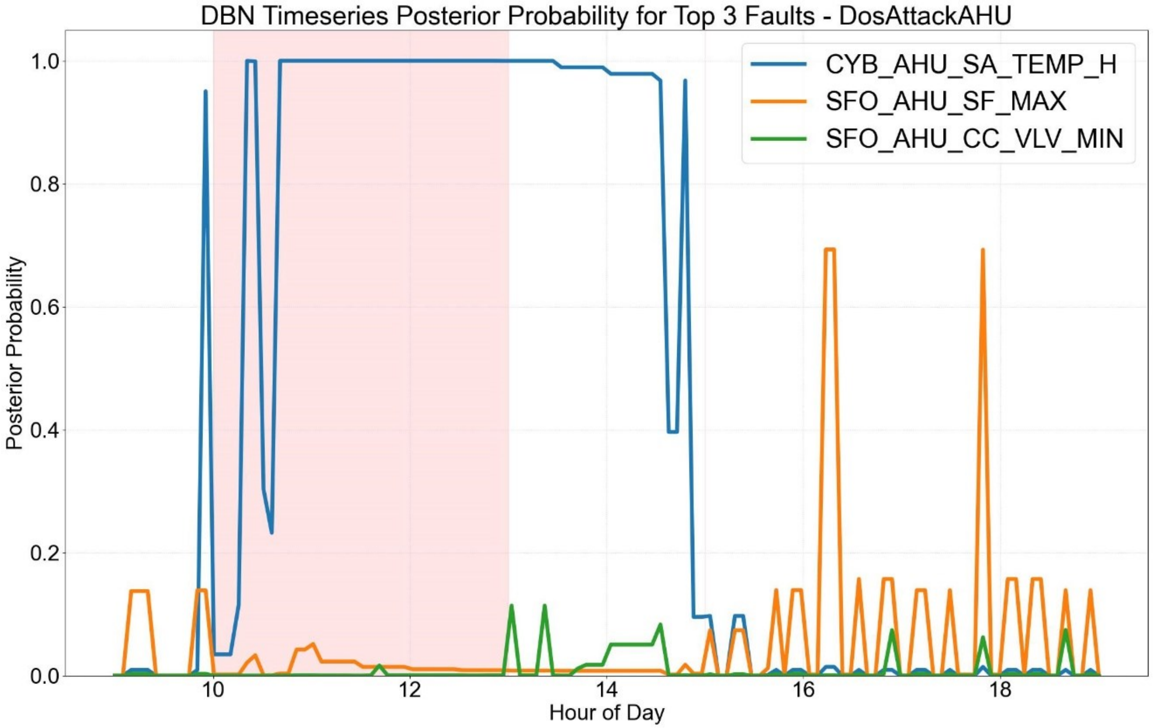


- AHU supply air temperature changed from 65°F to 55°F (10am to 2pm)
- System fully recovers around ~3pm

Task 2.3 Evaluation: HIL Testbed

Cyber Attack - Network DoS attack on AHU





- Post-attack system abnormalities when system is recovering <u>not identified</u> by static BN
- DBN characterizes physical measurements <u>more effectively</u>

Conclusions of Objective 2

- A discretized DBN framework is proposed to represent temporal relationships to track fault evolution
- DBN allows fault beliefs to accumulate over time resulting in improved diagnosis and successfully distinguishes cyber-attacks from physical faults
- CCIE-based approach more effective for constructing baseline using historical data

Future Tasks:

- Standardize method of evaluation using sample-level classification metric
- Integrate CCIE-based baseline construction approach with proposed DBN framework and perform additional tests at HIL testbed
- Explore transfer learning of DBN structure and parameters from virtual to real buildings

Objective 3: Development of Fault Prognosis Strategy for Degradation Faults

Task 3.1 Develop a DBN-based fault prognosis framework to estimate future fault probabilities

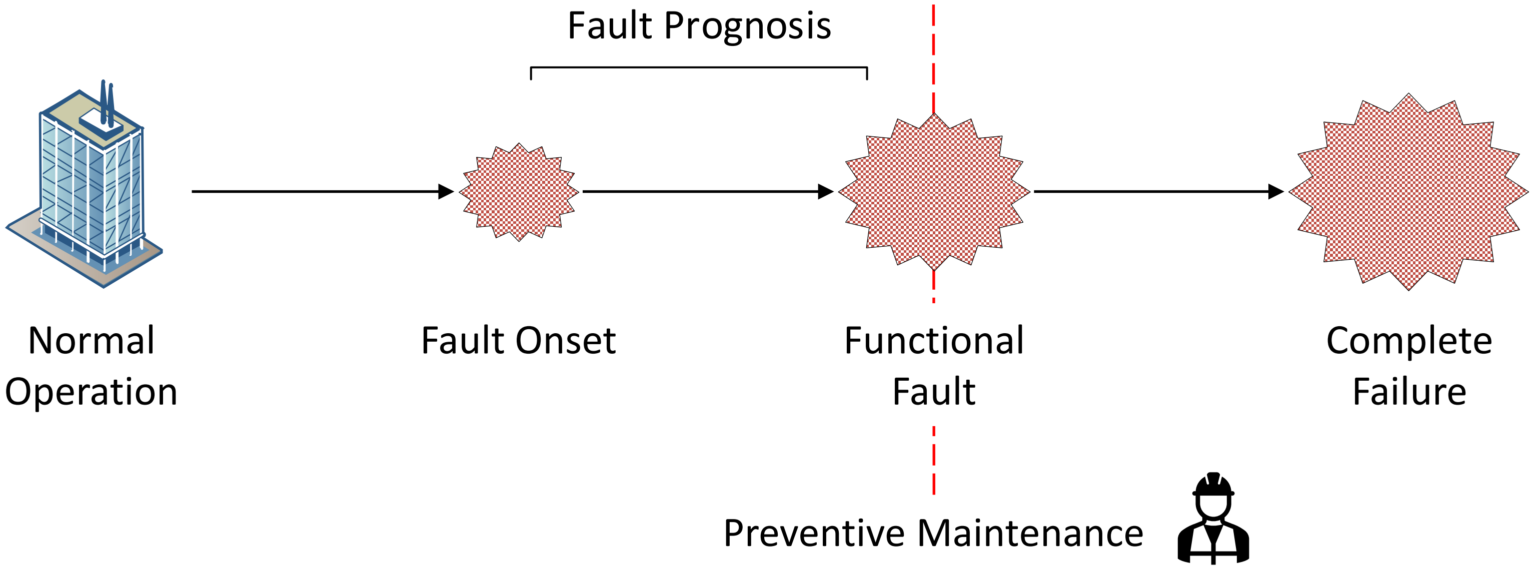
Task 3.2 Integrate a Robust Multivariate Temporal (RMT) variate selection process to identify key inputs for training forecasting models

Task 3.3 Evaluate the performance of the DBN-RMT framework using gradual fault data simulated in the virtual testbed

Task 3.1 Fault Prognosis

Introduction:

- Process of predicting future fault behaviors and estimating operation time before a failure occurs
- Enables active maintenance and servicing of HVAC components prior to their failure

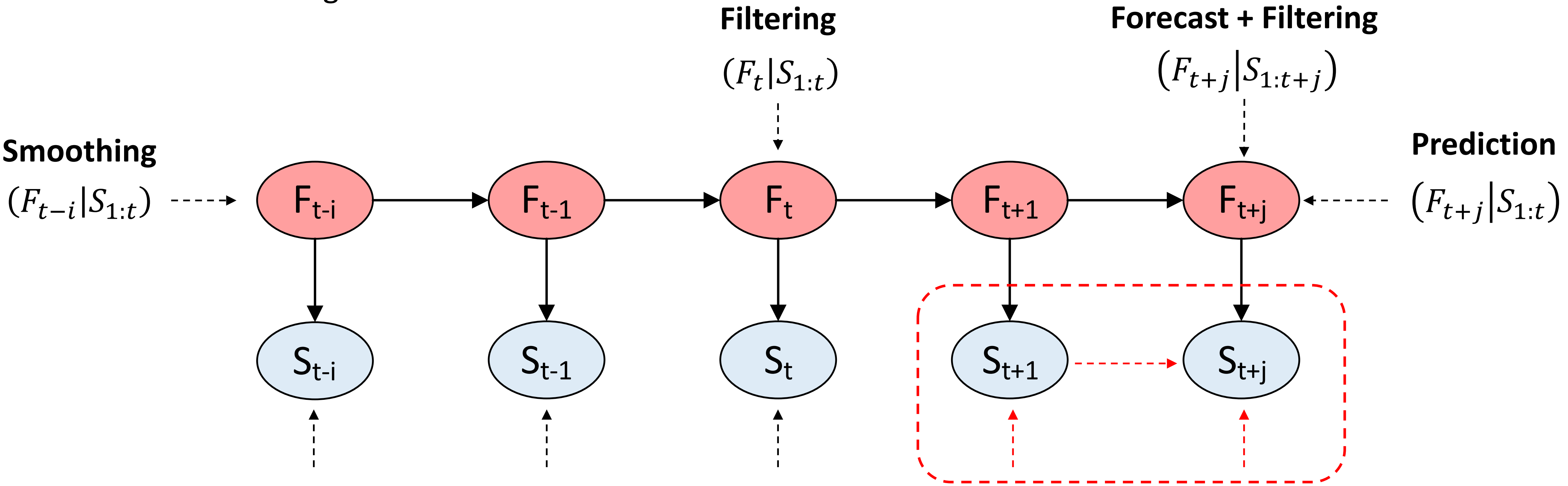


Challenges:

1. Prediction accuracy of future fault probabilities
2. High data dimensionality for forecasting models

Task 3.1 DBN Inferences

Potential of DBN for Prognosis



How to forecast $S_{t+1:t+j}$?

1. Temporal probabilities ← not feasible since measurements are dynamic
2. Forecasting models ← effective in learning trends and seasonality

Task 3.2 RMT Variate Selection

High data dimensionality <u>impedes</u> the development of data-driven forecasting models

<u>Objective:</u> find robust localized features along Gaussian scale and time to support variate selection

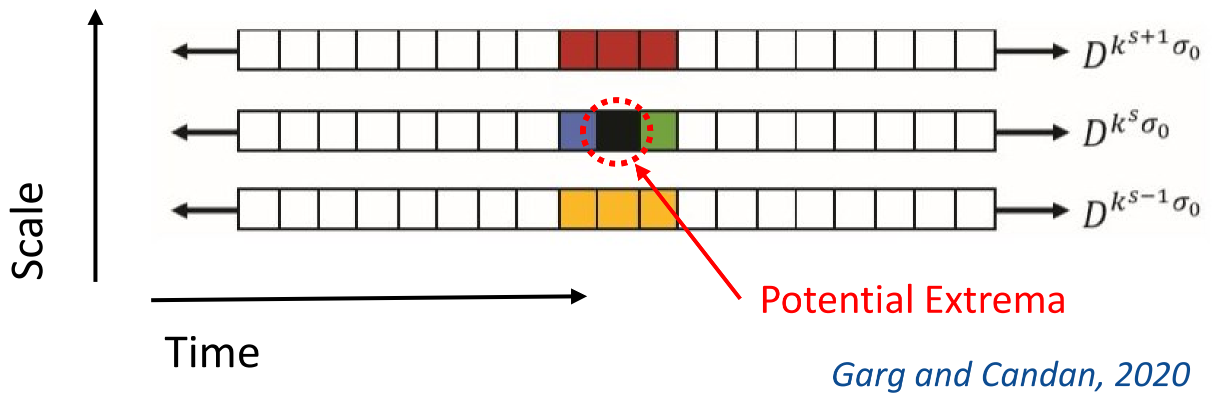
1. Create a scale-space consisting of multiple smoothed versions for a given variate

2. Search for local extrema across scale and time

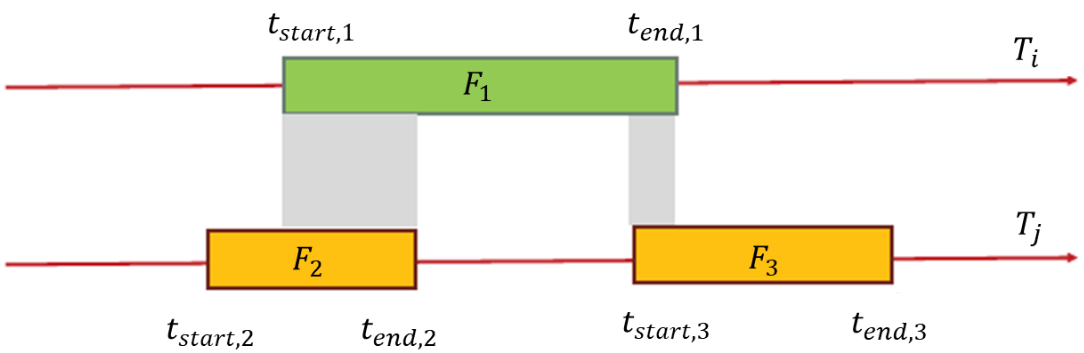
3. Calculate feature-to-feature alignments

4. Measure temporal alignment and generate variate ranking

Garg and Candan, 2020

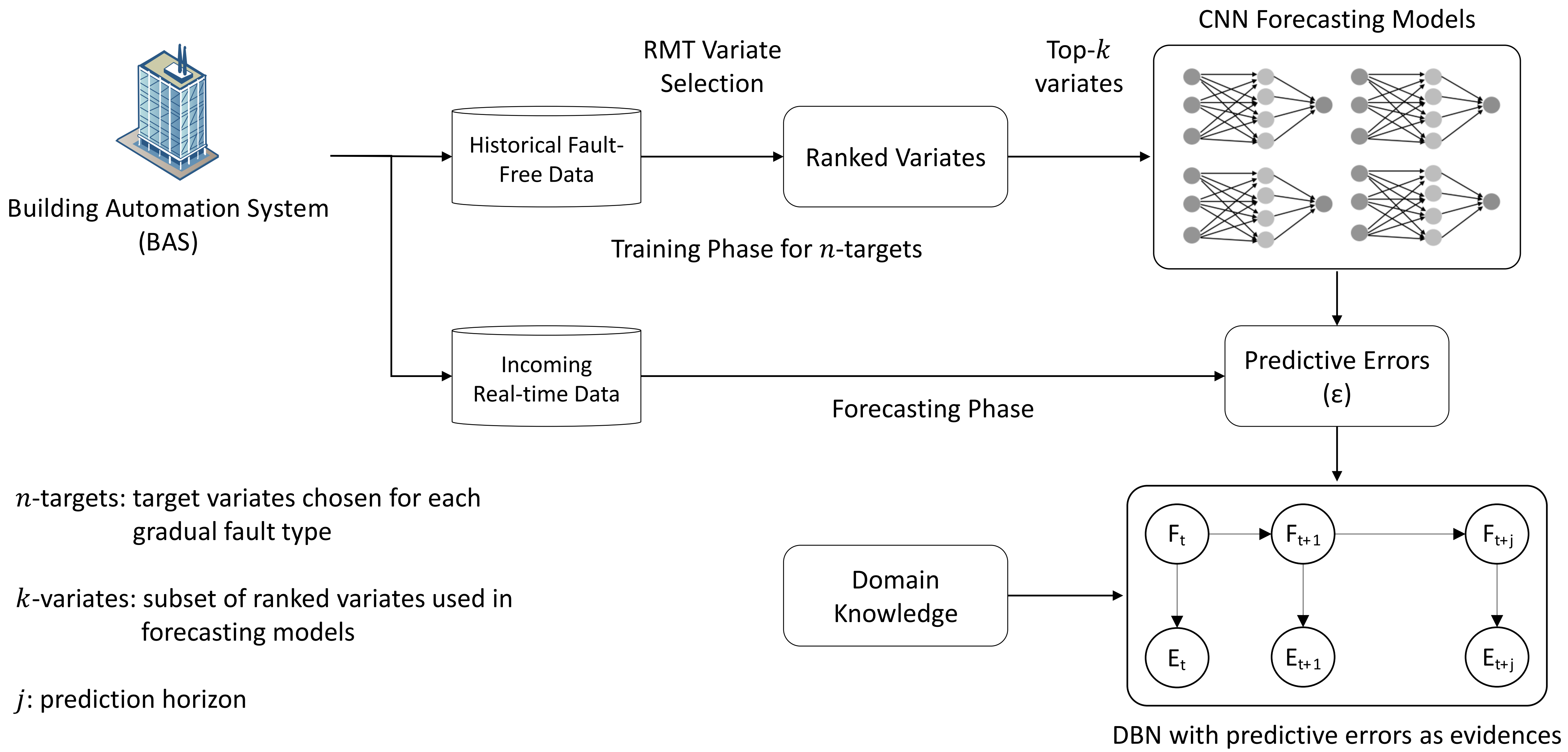


Garg and Candan, 2020



$$overlap(F_1, F_2) = min(t_{end,1}, t_{end,2}) - max(t_{start,1}, t_{start,2})$$

Task 3.3 DBN-RMT Framework



n-targets: target variates chosen for each gradual fault type

k-variates: subset of ranked variates used in forecasting models

j: prediction horizon

Task 3.3 Evaluation

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Simulated Gradual Faults *(Chu et al., 2022)*

Case	Case Description	Severity Level	Target Variate
1	Cooling coil airside fouling	Heat transfer coefficient decreases at a rate of 14%/year, pressure drop increases at a rate of 200%/year	AHU supply air temperature
2		Heat transfer coefficient decreases at a rate of 7%/year, pressure drop increases at a rate of 30%/year	
3	Condenser waterside fouling	Heat transfer rate decreases at a rate of 28%/30 days, Pressure drop increases at a rate of 50%/30 days	AHU supply air temperature
4		Heat transfer rate decreases at a rate of 7%/30 days, Pressure drop increases at a rate of 250%/30 days	
5	AHU supply air duct leakage	Starts at a 1% leakage, increases at a rate of 1.4%/year	AHU supply air flowrate
6		Starts at a 1% leakage, increases at a rate of 7%/year	
7		Starts at a 10% leakage, increases at a rate of 1.4%/year	
8		Starts at a 10% leakage, increases at a rate of 7%/year	
9		Starts at a 25% leakage, increases at a rate of 1.4%/year	
10		Starts at a 25% leakage, increases at a rate of 7%/year	
11	AHU supply air temperature sensor drifting	Sensor drifts at a rate of +1°C/year	AHU supply air temperature
12		Sensor drifts at a rate of -1°C/year	

Conclusions of Objective 3

- A DBN-RMT framework is proposed for fault prognosis:
  - DBN is used to characterize causal relationships, however, performs poorly in estimating future fault probabilities
  - CNN models can be utilized to forecast future fault symptoms together with DBN
  - RMT variate selection can rank variates for training forecasting models for high dimensional data

Future Tasks:

- Implement RMT variate selection for each target variate and train CNN forecasting models
- Evaluate overall DBN-RMT framework using gradual fault data collected from the virtual testbed
- Assess suitable integration approach with fault diagnosis

Overall Conclusions and Future Works

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Conclusions	Future Works
Objective 1: Reported data imputation methods suitable for handling and repairing BAS data are evaluated	
○ Incorporating time-lagged cross correlations in kNN method significantly improves imputation accuracy	○ Evaluate impact on information entropy and fault detection
Objective 2: A systematic and scalable DBN framework is proposed for cyber-physical fault diagnosis in GEBs	
○ Uses discretized conditional probabilities to track fault evolution ○ DBN more accurate and representative of the observed symptoms ○ CCIE-based approach both constructs and evaluates baseline from historical data	○ Evaluate using standardize sample-level classification ○ Integrate CCIE-based approach with DBN framework ○ Explore DBN learning algorithms
Objective 3: A DBN-RMT framework is proposed for fault prognosis for degradation faults	
○ DBN represents fault-symptom causal relationships ○ RMT variate selection determines best inputs to forecast future fault probabilities	○ Develop and evaluate the framework using gradual fault data

Overall Conclusions and Future Works

Future Work Schedule

Task	Task Description	Jul-2022	Aug-2022	Sep-2022	Oct-2022	Nov-2022	Dec-2022	Jan-2023	Feb-2023	Mar-2023
1	**Data Quality and Data Imputation**									
1.1	Investigate additional details of lagged-kNN method									
1.2	Evaluate impacts of data imputation methods on fault detection and diagnosis algorithms									
1.3	Journal paper #1 writing						◇			
2	**Dynamic Bayesian Network for Fault Diagnosis**									
2.1	Standardize method evaluation for all datasets using joint classification scheme									
2.2	Integrate DBN with CCIE method and perform additional tests using HIL dataset									
2.3	Journal paper #2 writing				◇					
3	**Dynamic Bayesian Network for Fault Prognosis**									
3.1	Investigate data-driven methods for learning DBN parameters									
3.2	Develop DBN-RMT based fault prognosis strategy for gradual faults									
3.3	Evaluate developed method using data from virtual testbed									
3.4	Journal paper #3 writing							◇		
4	**Thesis Writing**									
4.1	Finish thesis writing									

◇ *Journal publication milestone*

Overall Conclusions and Future Works

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Related Publications (4 conference proceedings and 1 journal publication):

1. Pradhan, O., Hälleberg, D., Chen, Z., Wen, J., Wu, T., Candan, K. S., & O'Neill, Z (2022). Lagged-kNN Based Data Imputation Approach for Multi-Stream Building Systems Data. 7th International High-Performance Buildings Conference at Purdue. July 10-14, 2022. (Accepted)

2. Pradhan, O., Wen, J., Chen, Y., Lu, X., Chu, M., Fu, Y., O'Neill, Z, Wu, T., & Candan, K. S. (2021). Dynamic Bayesian Network-based Fault Diagnosis for ASHRAE Guideline 36: high performance sequence of operation for HVAC systems. *In Proceedings of the 8th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (pp. 365-368).*

3. Pradhan, O., Wen, J., Chen, Y., Wu, T., & O'Neill, Z. (2021). Dynamic Bayesian Network for Fault Diagnosis. *2021 ASHRAE Virtual Conference.* June 28 – June 30, 2021.

4. Huang, J., Wu, T., Yoon, H., Pradhan, O., Wen, J., & O'Neill, Z. (2021). Automatic Fault Detection Baseline Construction for Building HVAC Systems using Joint Entropy and Enthalpy. *In IIE Annual Conference Proceedings (pp. 536- 541).*

5. Huang, J., Yoon, H., Pradhan, O., Wu, T., Wen, J., O'Neill, Z., & Candan, K. S. (2022). A Cosine-based Correlation Information Entropy Approach for Building Automatic Fault Detection Baseline Construction. *Science and Technology for the Built Environment.*

Thank You

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Backup

Overview of Datasets and Testbeds

FHS Building (Stockholm, Sweden)

- Mixed-use institution building
- Multiple data sources
- District heating, cooling and electric supply
- Used for Objective 1

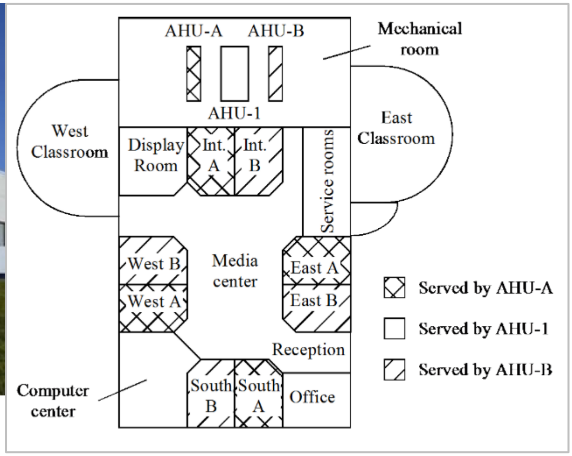
Nesbitt Hall (Drexel University)

- Mixed-use institution building
- Water-cooled chillers
- Multi-zone VAV
- Used for Objective 2
  - Normal (fault-free) operation data
  - 14 artificially injected fault cases

Chen 2018; Zhang 2018

Laboratory Test Facility (ASHRAE RP-1312)

- Small commercial building
- Twin AHU system (AHU-A and B)
- Used for Objective 2
  - AHU-B running in fault-free operation
  - 11 artificially injected fault cases



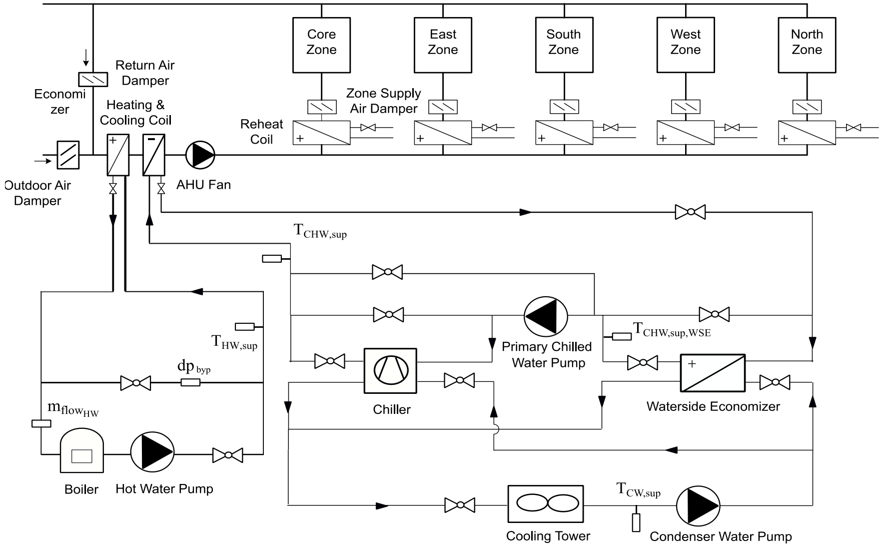
ERS Test Facility
Wen & Li, 2011

Overview of Datasets and Testbeds

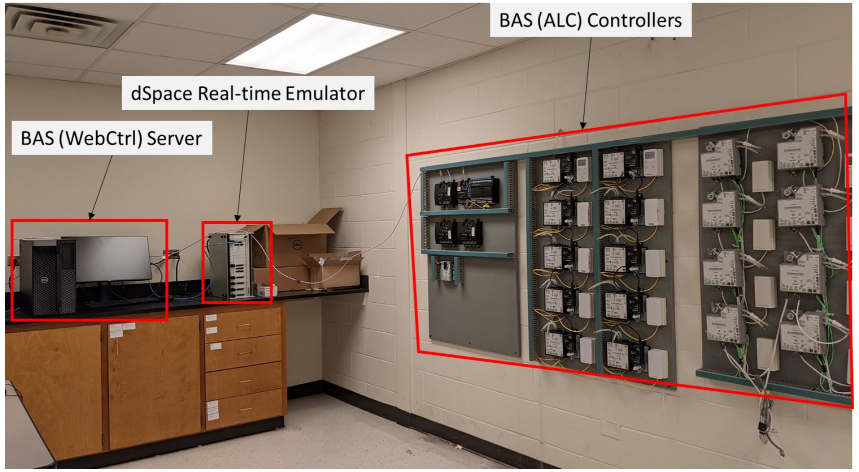
Virtual Testbed: Medium Office Building

- Based on DOE reference medium-sized building
- One AHU connected with 5 VAV terminals serving 5 zones
- ASHRAE's Guideline 36 control sequence *(ASHRAE 2018)*
- Integrated with Hardware-in-the-Loop (HIL) testbed for real-time emulation

Dataset	Fault Configuration	Fault Cases	Used for Objective
MedOffice-Phy	Simulated Physical Faults (Instantaneous)	13	2
MedOffice-Cyb	Simulated Cyber Attacks	4	
MedOffice-PhyHIL	Physical Faults Injected in HIL Testbed	2	
MedOffice-CybHIL	Cyber Attacks Injected in HIL Testbed	1	
MedOffice-Gradual	Simulated Physical Faults (Gradual)	12	3



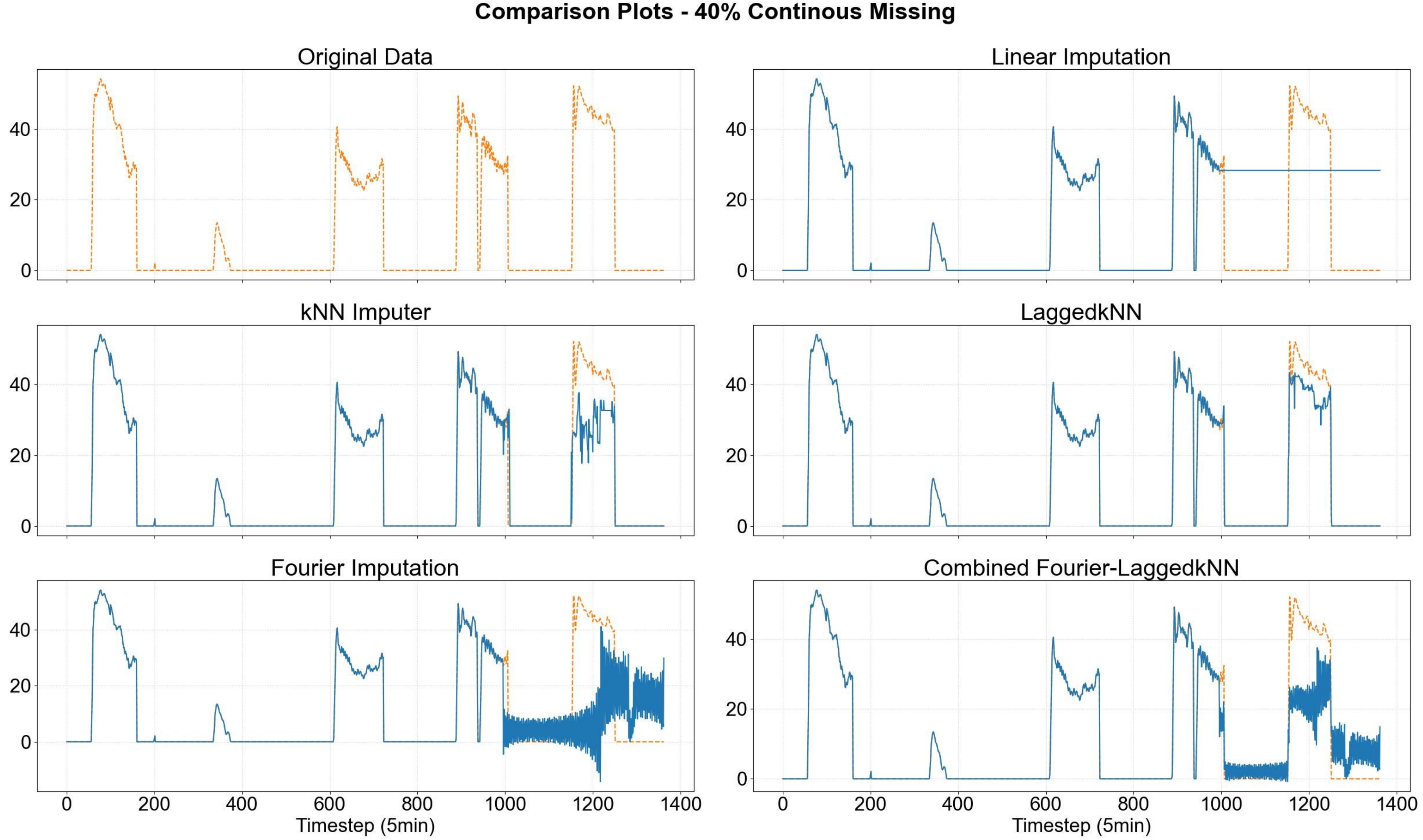
Virtual building model in Modelica
Fu et al., 2021; Lu et al., 2021



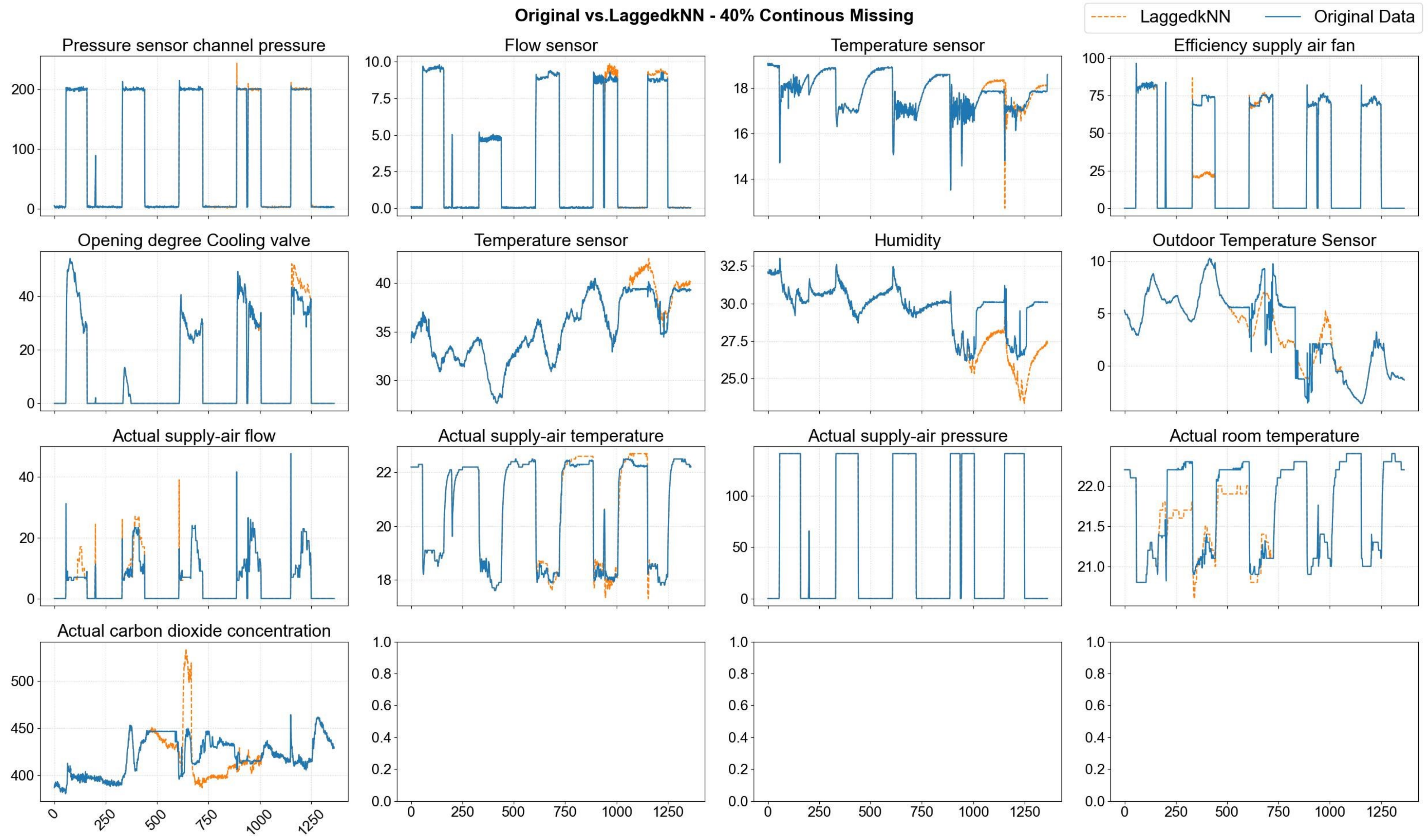
HIL testbed at TAMU

Task 1.2 & 1.3 Evaluation

AHU Cooling Coil Valve Position (40% Continuously Missing)



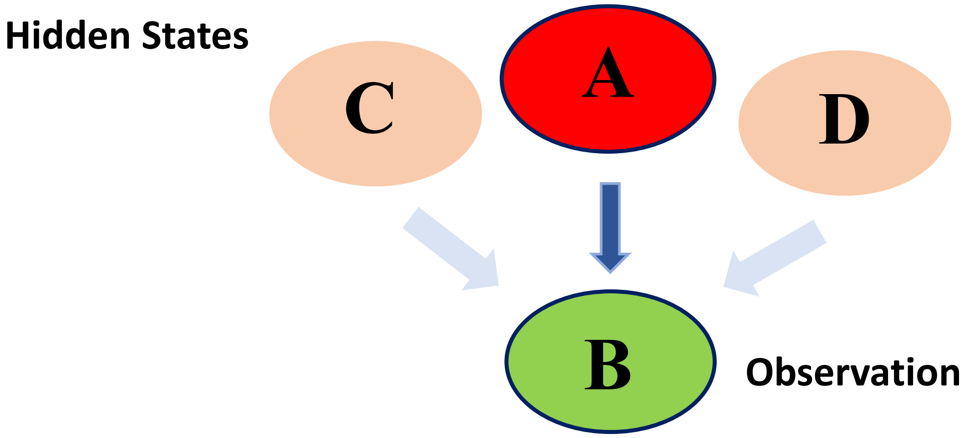
Task 1.2 & 1.3 Evaluation



Task 2.1 Bayesian Networks

Bayesian Networks (BNs): A probabilistic model which reveal causal relations between faults and symptoms [8]

- BNs can add uncertainty factors into the inference
- BNs provide a ranked list of the potential causes of faults



Bayesian Network Structure

Bayes' Rule:

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

$$P(B) = \sum_{i=1}^{n} P(A_i)P(B|A_i)$$

$P(A|B) - posterior\ probability$

$P(B|A) \ - conditional\ probability$

$P(A) - prior\ probability$

Task 2.1 Bayesian Networks

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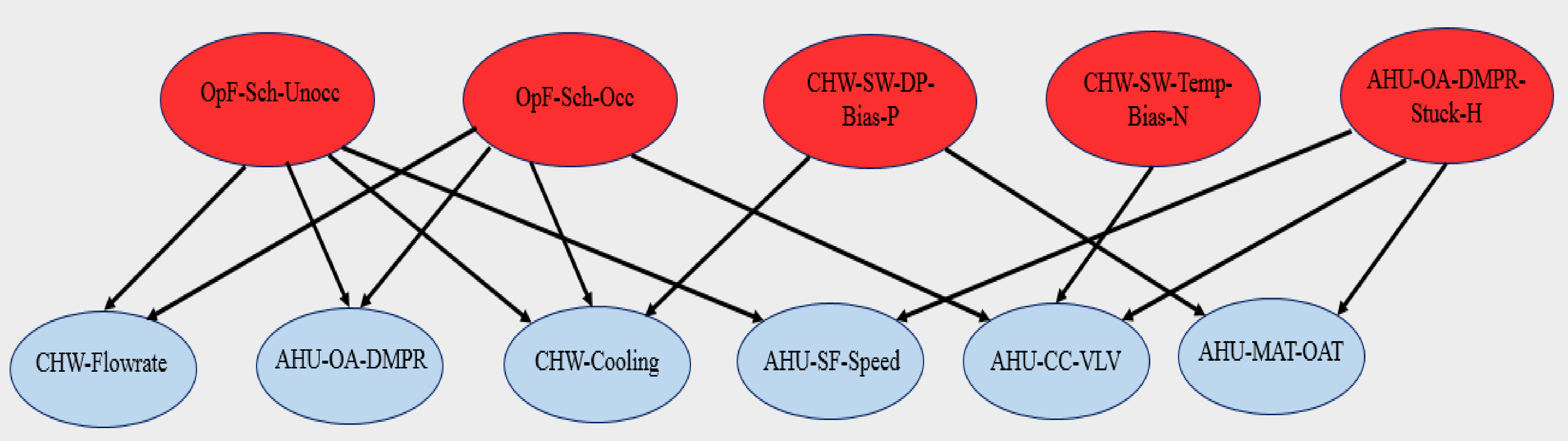
BN Structure Model:

- Two-layer BN structure with fault and evidence nodes
- Software architecture: Python + pySMILE library



Snapshot of Two-layer BN Structure
(Chen 2018)

Task 2.1 DBN Overview

Dynamic Bayesian Networks (DBN):

- Fault nodes at t have additional dependency on the fault node at $(t-1)$
- Temporal conditional probability (CPTs): probability of a fault event under the occurrence of a fault event at from the previous time step
- Temporal CPTs are defined as *(Murphy 2002)*:

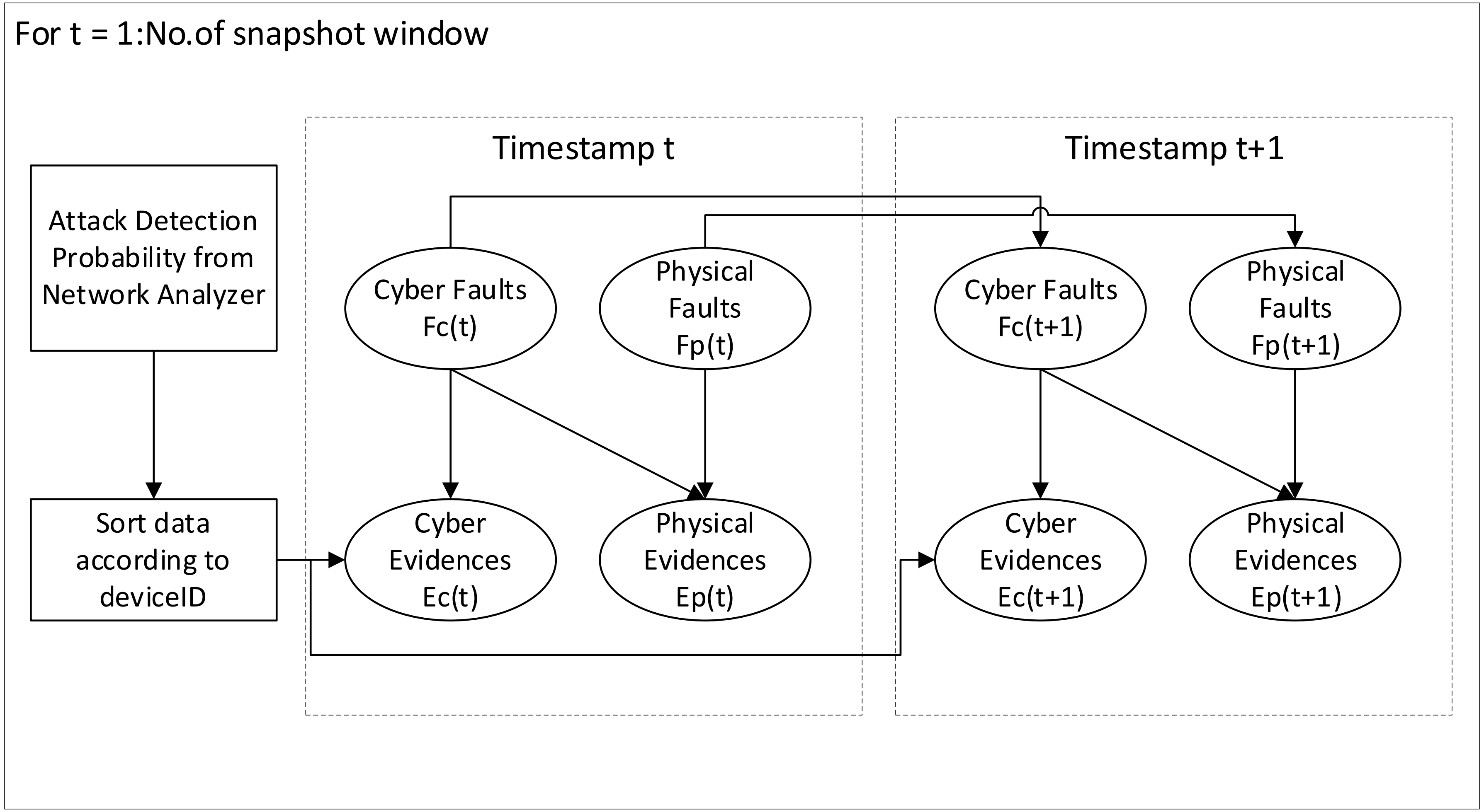
$$P(F_{t+1}|F) = \prod_{i=1}^{n} P\left(F_{i,t+1}|Pa\left(F_{i,t+1}\right)\right)$$

where $P\left(F_{i,t+1}\right)$ is the i^{th} node at time $t+1$, $Pa\left(F_{i,t+1}\right)$ is the parent nodes of $F_{i,t}$ from same and previous time step, and n is the number of nodes in the network.

Task 2.1 DBN Framework

Step 1: Gather incoming snapshot and baseline data
↓
Step 2: Generate DBN structure model*
↓
Step 3: Define DBN parameters
↓
Step 4: Generate LEAK distribution
↓
Step 5: Generate DBN evidence
↓
Step 6: Implement DBN inference*
↓
Step 7: Fault diagnosis and isolation

**Executed in pySMILE library*



Task 2.1 Causal Relations Example

Fault No.	Fault Node Description	Evidence Node Name	Evidence Node Description	Evidence Direction	Evidence Type
1	AHU Outdoor Air Damper Stuck Higher than Normal	CHWC_VLV	Cooling coil valve position	Positive	Moderate
		MA_TEMP_1	Vs. MAT = f(OAT, RAT, SAcfm, RAcfm)	Negative	Strong
		MA_TEMP_2	Vs. MAT = f(OAT, RAT,Sacfm,OAdmpr)	Positive	Strong
		CHW_Flow	Chilled water flowrate	Positive	Moderate
		CHW_Cooling	Chilled water cooling	Positive	Moderate
		OA_CFM	Outdoor air flowrate	Negative	Strong
2	AHU Outdoor Air Damper Stuck Lower than Normal	CHWC_VLV	Cooling coil valve position	Negative	Weak
		MA_TEMP_1	Vs. MAT = f(OAT, RAT, SAcfm, RAcfm)	Positive	Strong
		MA_TEMP_2	Vs. MAT = f(OAT, RAT,Sacfm,OAdmpr)	Negative	Strong
		CHW_Flow	Chilled water flowrate	Negative	Light
		CHW_Cooling	Chilled water cooling	Negative	Moderate
		OA_CFM	Outdoor air flowrate	Negative	Strong
		OA_DMPR	Outdoor air damper	Positive	Strong
3	AHU Cooling Coil Valve Stuck Higher than Normal	CHWC_VLV	Cooling coil valve position	Negative	Strong
		DIFF_SAT_STP	Difference between supply air temperature and supply air setpoint temperature	Negative	Strong
		CHW_Flow	Chilled water flowrate	Positive	Strong
		HWC_VLV	Heating coil valve position	Positive	Strong
		CHW_Cooling	Chilled water cooling	Positive	Strong
4	AHU Cooling Coil Valve Stuck Lower than Normal	CHWC_VLV	Cooling coil valve position	Positive	Strong
		DIFF_SAT_STP	Difference between supply air temperature and supply air setpoint temperature	Positive	Strong
		CHW_Flow	Chilled water flowrate	Negative	Strong
		CHW_Cooling	Chilled water cooling	Negative	Strong

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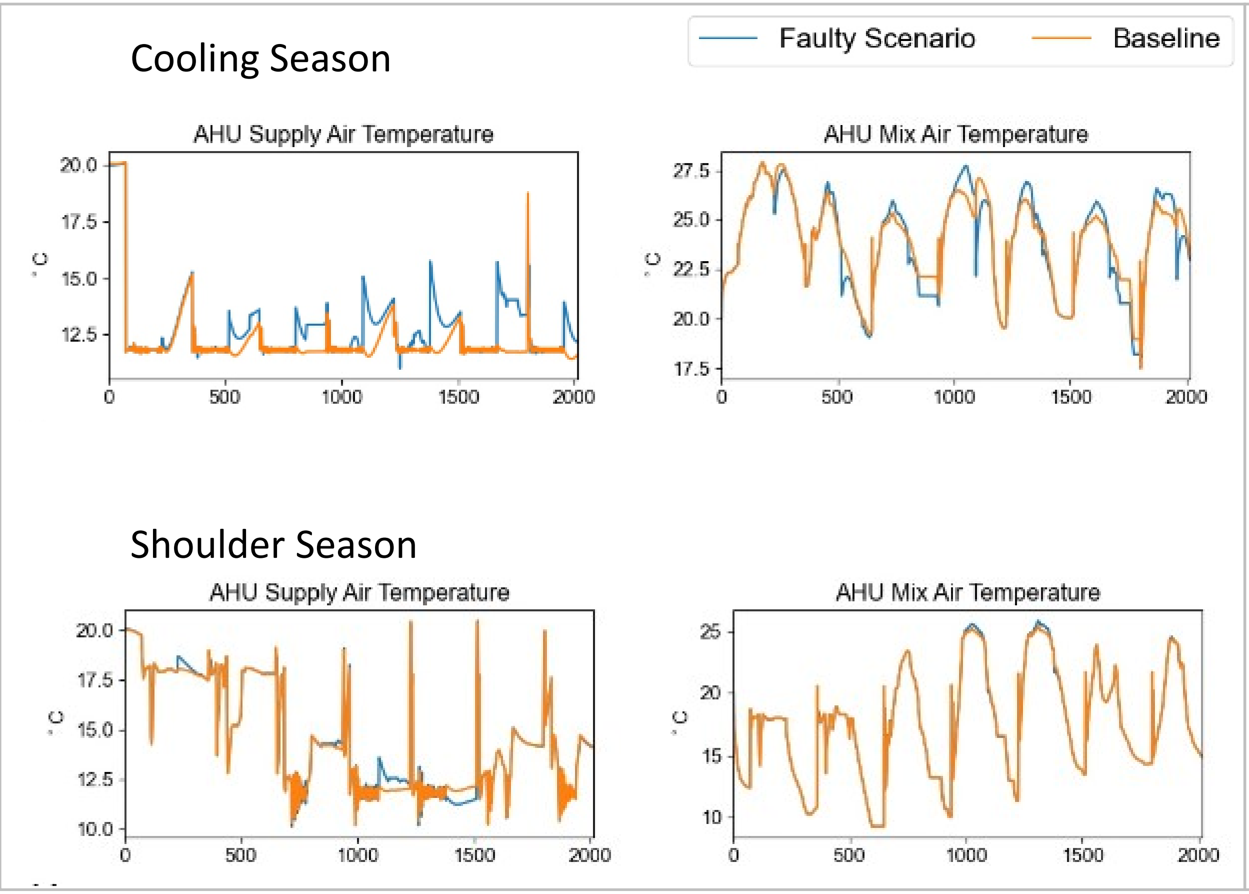
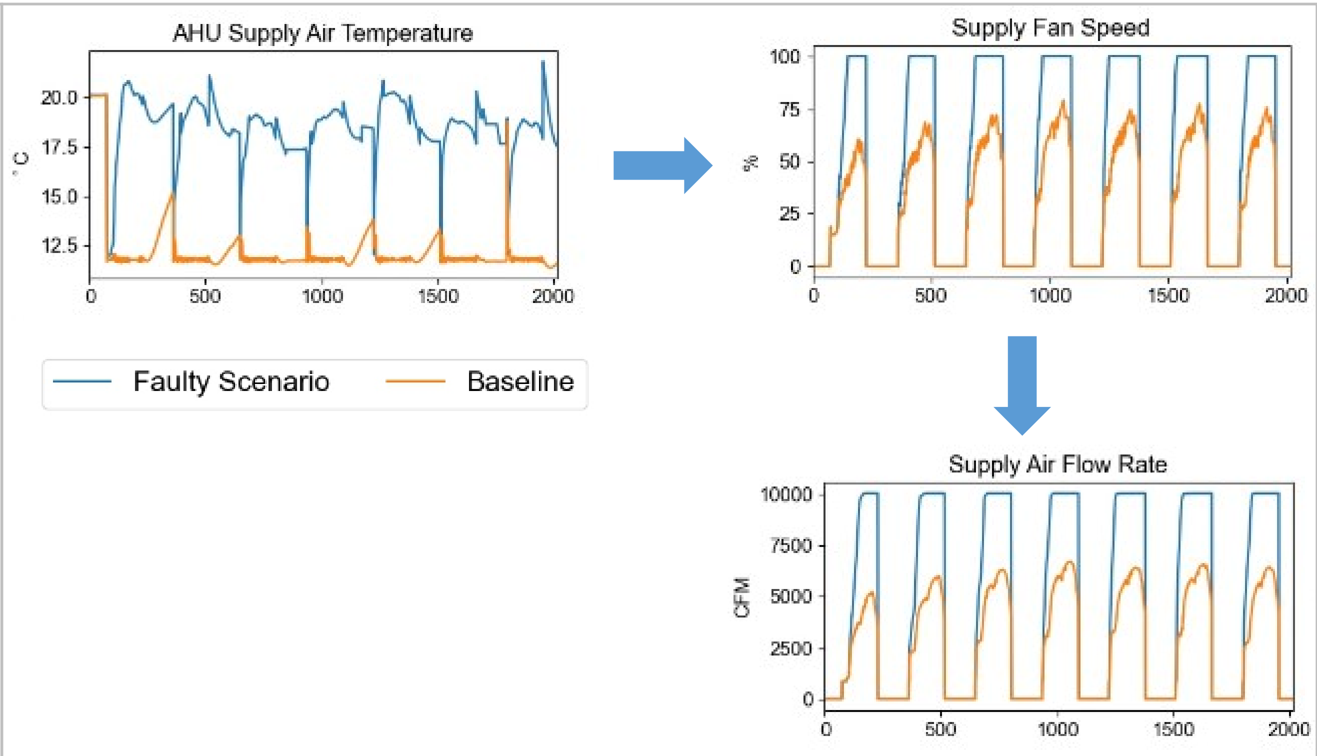
Limitations of Knowledge-based Causal Relations



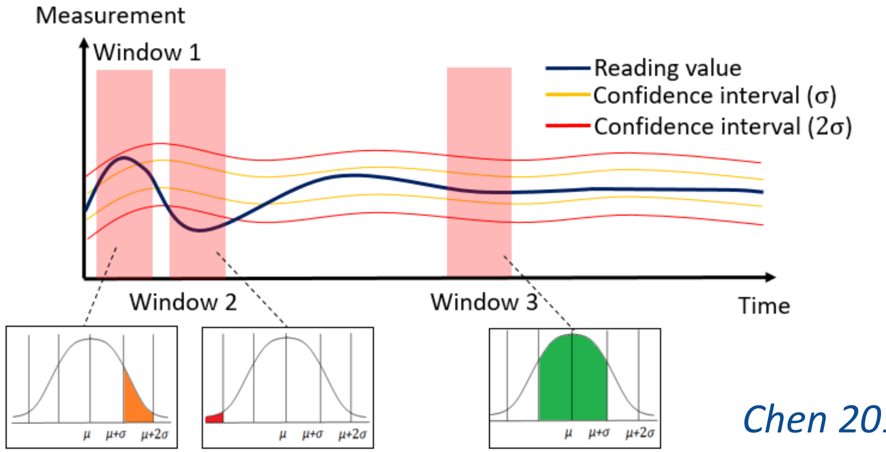
AHU supply air and mixed air temperatures during AHU outdoor air stuck at 100% fault



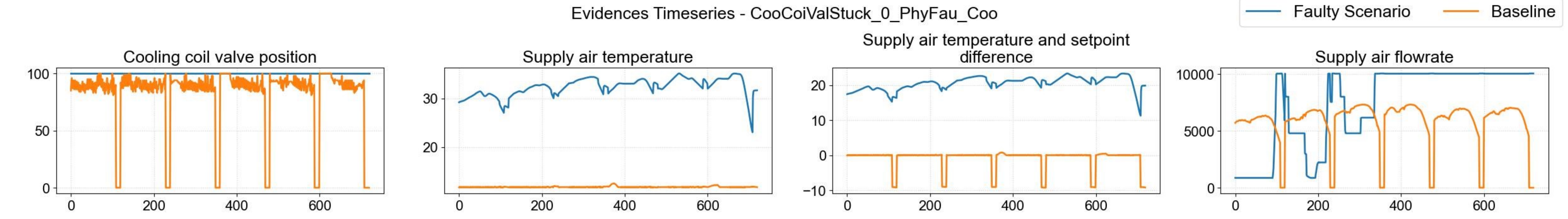
Downstream impact of the AHU cooling coil valve high fault

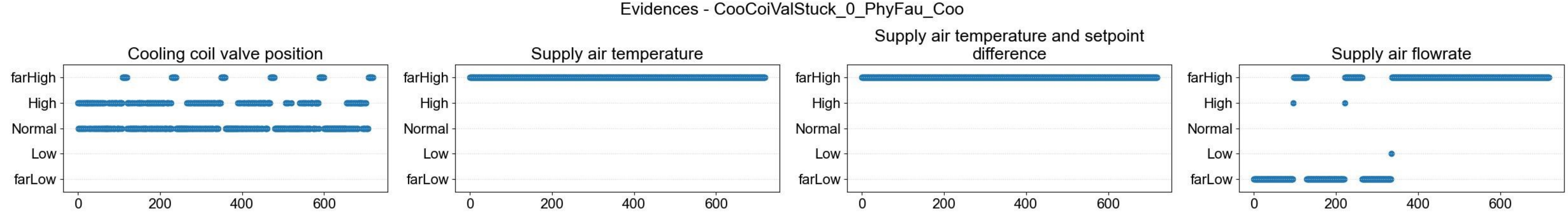
Task 2.1 Evidence Generation

- Normal or Fault-Free: within 1σ
- Moderate Evidence: between 1σ and 2σ
- Strong Evidence: over 2σ



Chen 2018





Task 2.1 Fault Ranking

The following rules are defined to classify if a fault is diagnosed, i.e., identified as the root-cause for an abnormality:

- the posterior probability of first ranking fault node is higher than 15% and
- the posterior probability of this fault node is at least 10% higher than the second-highest
- Alternatively, a case is identified as fault free if the posterior probability of first ranking fault node is lower than 15%.

$$P_{all}(A) = \frac{\sum_{i=1}^{n} P_i(A)}{n}$$

$P_{all}(A)$ is the calculated posterior probability during faulty period

$P_i(A)$ is the calculated posterior probability based on each incoming data sample

n is the total number of samples in the faulty period.

Task 2.1 Fault Ranking

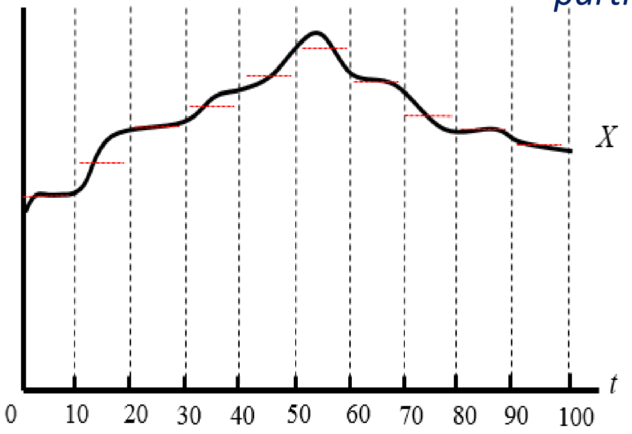
Class	Network Analyzer Detection	PCA Fault Detection	DBN Fault Diagnosis	Outcome
0	-	-	-	No cyber-attacks nor physical system faults
1	Yes	Yes	Yes	Cyber-attack causing physical system abnormality
2	-	Yes	Yes	Physical faults causing physical system abnormality
3	Yes	Yes	-	Cyber-attack causing physical system abnormality; not diagnosed
4	Yes	-	-	Only cyber-attacks; no physical system abnormality

$$Accuracy = \frac{Number\ of\ correctly\ classified\ samples}{Total\ number\ of\ samples\ classified}$$

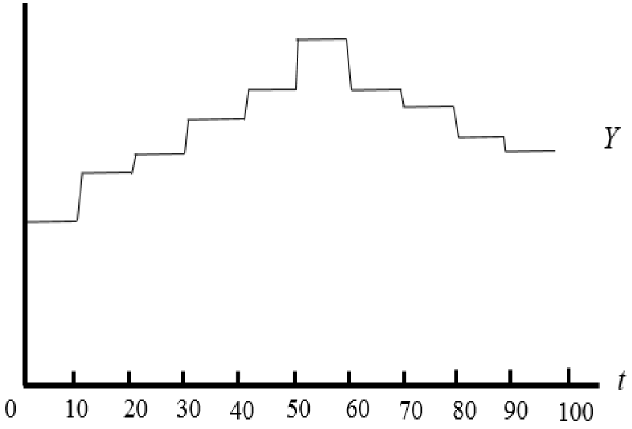
$$False\ Positive\ Rate\ (FPR) = \frac{Number\ of\ incorrectly\ classified\ samples}{Total\ number\ of\ samples\ outside\ the\ true\ class}$$

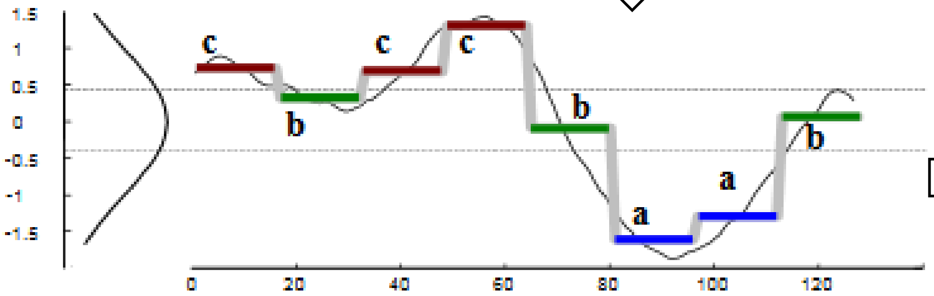
Symbolic Aggregate ApproXimation (SAX) method *(Chen 2018)*

PAA



Time series data is partitioned



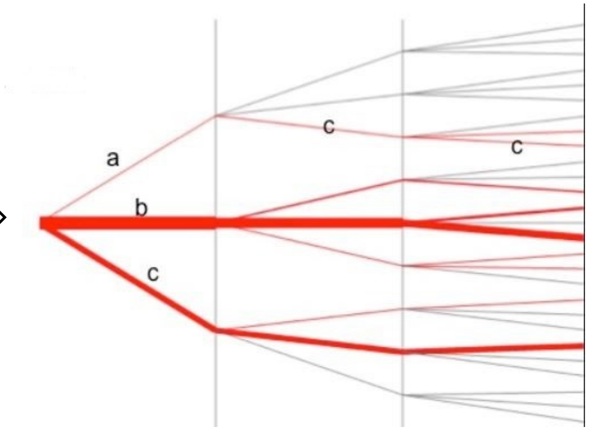


Time series data is tagged

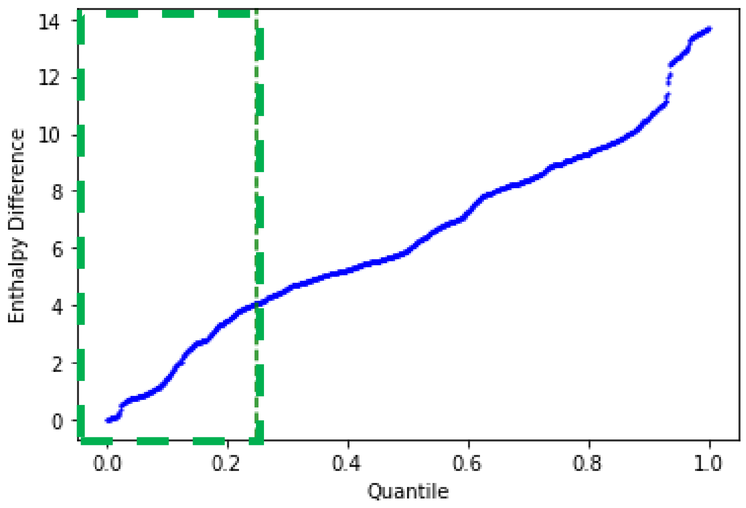
SAX

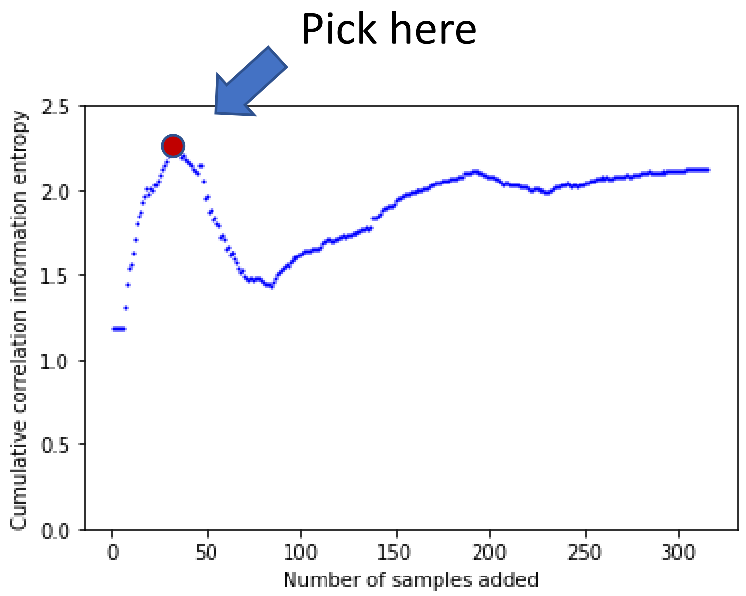
cbccbaab

Clustering



Cosine Correlation Information Entropy (CCIE) approach *(Huang et al., 2022)*





Phase I: Enthalpy is used for selecting samples to construct candidate pool

Phase II: Entropy is used to determine final baseline

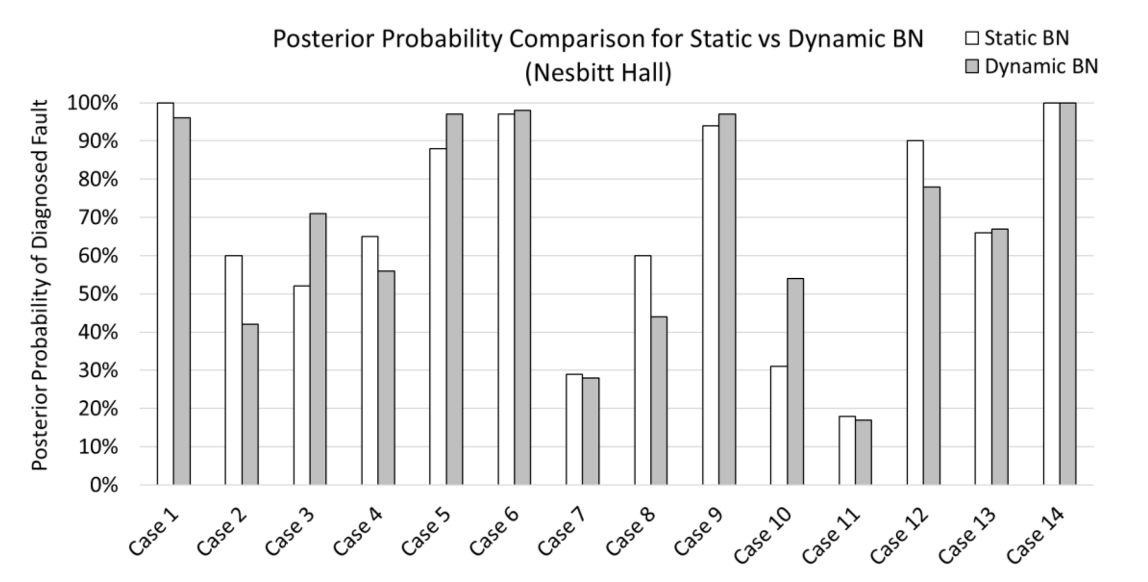
Task 2.3 Evaluation

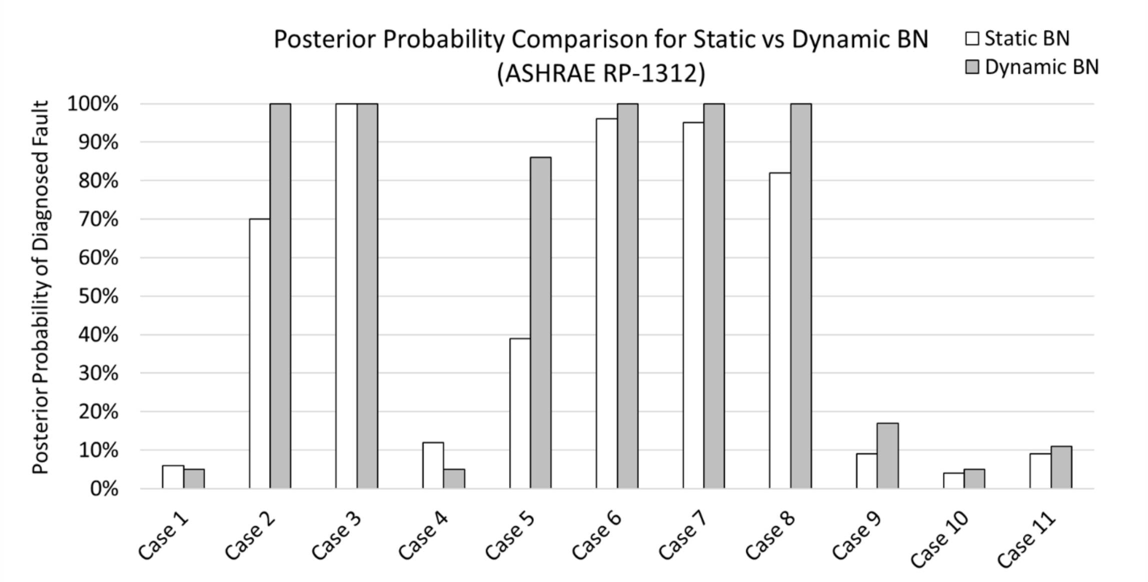
- Fault diagnosis result using BN and DBN in Nesbitt Hall (using averaged fault belief):

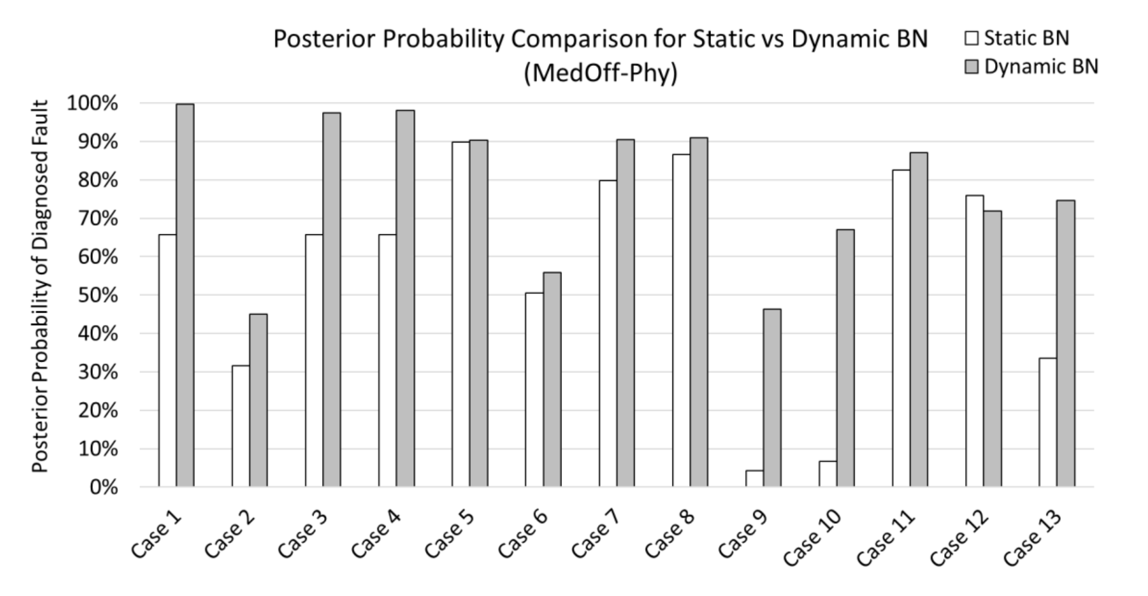
Date	Fault Description	BN Diagnosis Result	DBN Diagnosis Result
7/6/2016	Chiller stopped earlier	Diagnosed	Diagnosed
8/8/2016	AHU-1 supply air temperature sensor negative bias 4°F	Diagnosed	Diagnosed
9/7/2016	AHU-2 supply air temperature sensor negative bias 4°F	Diagnosed	Diagnosed
9/11/2016	Operator fault, chiller off	Diagnosed	Diagnosed
7/11/2017	AHU-2 outdoor air damper stuck at 90% open (higher than normal)	Diagnosed	Diagnosed
7/18/2017	AHU-2 outdoor air damper stuck at 100% open (higher than normal)	Diagnosed	Diagnosed
8/3/2017	Chiller chilled water supply temperature sensor negative bias 4°F	No fault symptoms	No fault symptoms
8/11/2017	AHU-2 cooling coil valve position software override at 100% open (higher than normal)	Diagnosed	Diagnosed
9/15/2017	Chiller chilled water differential pressure sensor positive bias 0.1 psi	Diagnosed	Diagnosed
7/9/2018	AHU-2 supply air temperature sensor negative bias 3.5°F	Simultaneous fault	Simultaneous fault
7/10/2018	AHU-2 outdoor air damper stuck at 30% open (higher than normal)	Simultaneous fault	Simultaneous fault
7/11/2018	AHU-2 cooling coil valve stuck at 80% open (higher than normal)	Diagnosed	Diagnosed
7/18/2018	AHU-2 outdoor air damper stuck at 60% open (higher than normal)	Diagnosed	Diagnosed
7/22/2018	Changed occupied schedule to end early	Diagnosed	Diagnosed

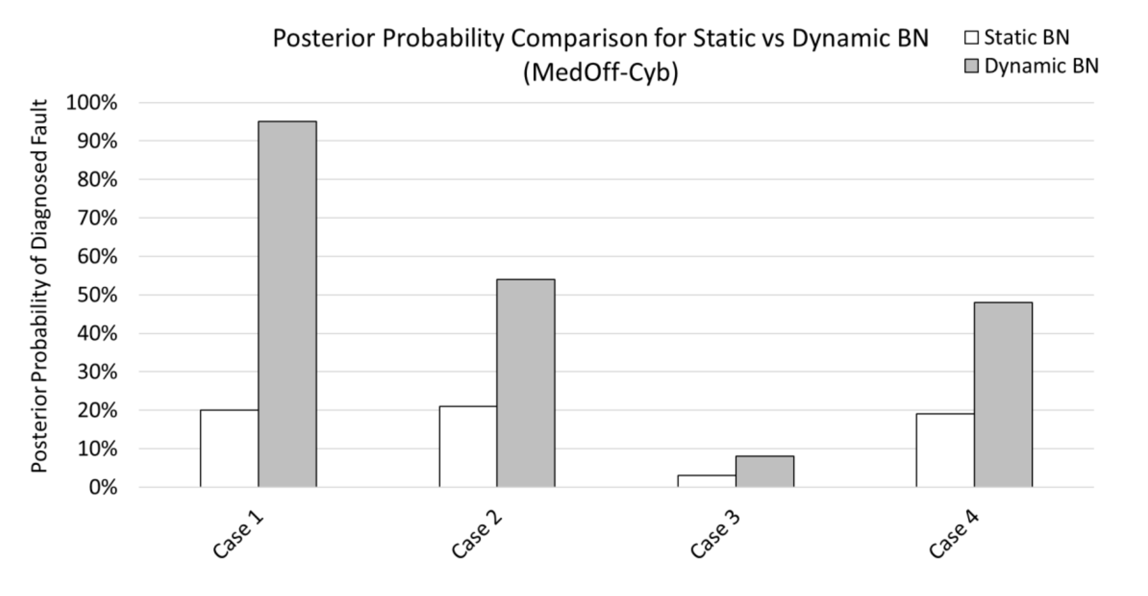
- Both methods diagnosed 11 out of 14 cases
- 2 cases where ground truth was unclear; 1 case where implemented fault showed no symptom

Task 2.3 Evaluation



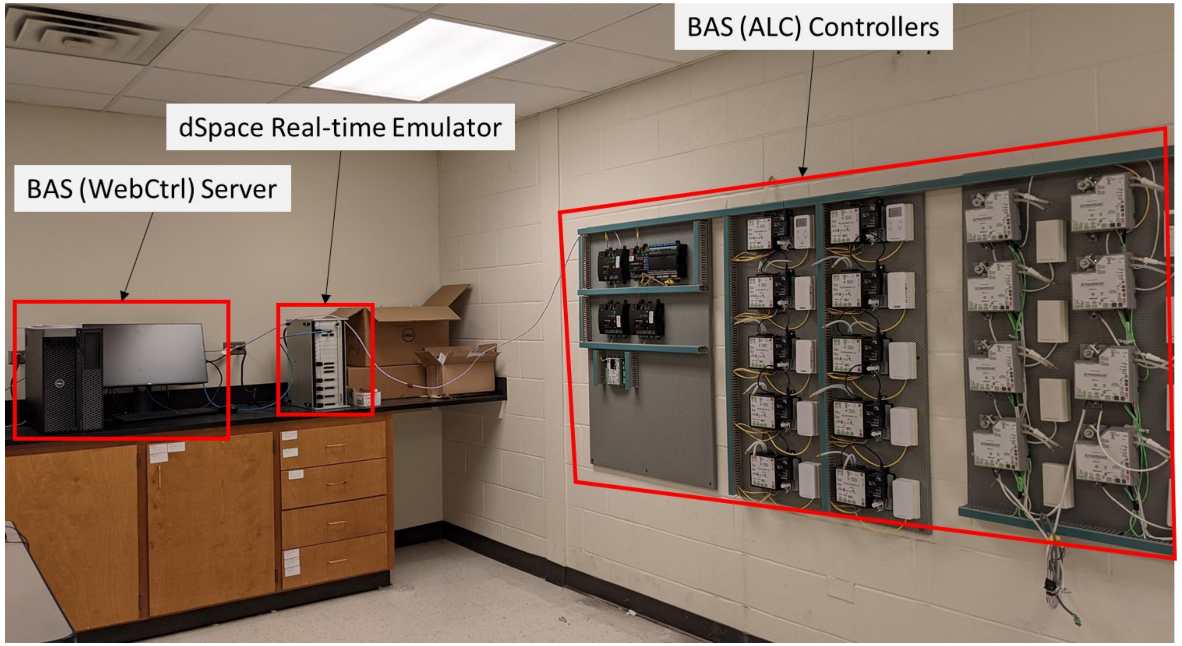




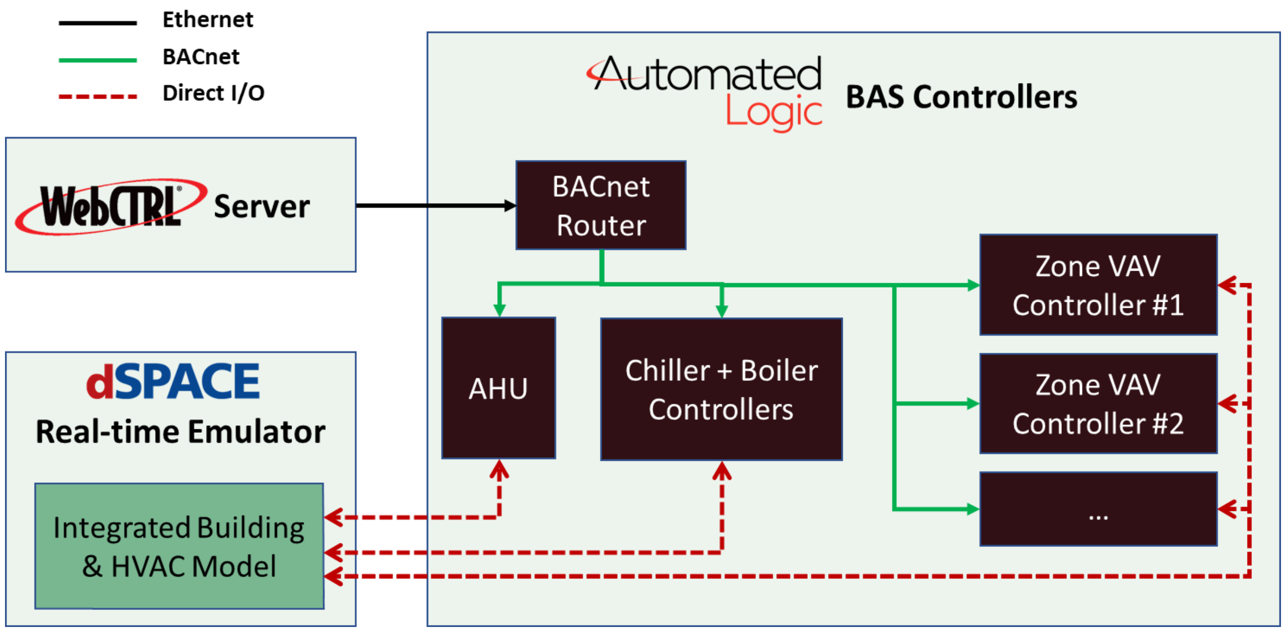


Task 2.3 Evaluation

- ALC-WebCTRL server computer → Supervise BAS operation & record data
- ALC-BAS router and equipment controllers → Execute control logic for typical HVAC equipment
- dSpace HIL real-time emulator → Emulate behaviors of building and HVAC equipment via simulation



HIL testbed at TAMU



Data flow in the HIL testbed

Task 2.3 Evaluation: HIL Testbed

Joint Classification Metric

Class	Network Analyzer Detection	PCA Fault Detection	DBN Fault Diagnosis	Outcome
0	-	-	-	No cyber-attacks nor physical system faults
1	Yes	Yes	Yes	Cyber-attack causing physical system abnormality
2	-	Yes	Yes	Physical faults causing physical system abnormality
3	Yes	Yes	-	Cyber-attack causing physical system abnormality; not diagnosed
4	Yes	-	-	Only cyber-attacks; no physical system abnormality

Dataset	Attack Type	Attack Scenario	Overall Accuracy	False Positive Rate
MedOff-PhyHIL	Fault injection	Cooling coil valve at minimum position	99%	<1%
MedOff-PhyHIL	Fault injection	AHU supply air temperature at maximum value	97%	4%
MedOff-CybHIL	Cyber-attack	Network DoS attack on AHU	95%	7%

Task 2.3 Evaluation: Sensitivity Test

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Sensitivity Test on Temporal Conditional Probabilities

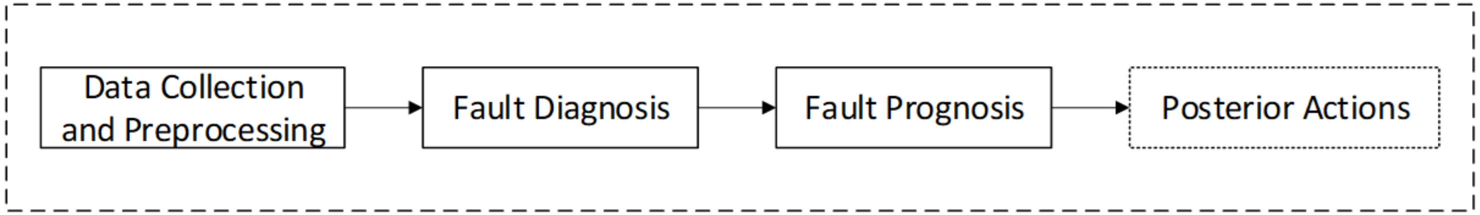
Test Configurations	Temporal Conditional Probability		
	Fault Node State (t-1)	**Fault Probability**	**Fault Free Probability**
Base Configuration	Fault Fault free	0.99 0.01	0.01 0.99
Test Configuration 1	Fault Fault free	0.8 0.2	0.2 0.8
Test Configuration 2	Fault Fault free	0.4 0.6	0.6 0.4

- Fault isolation outcome is not affected and does not change the fault ranking
- Changing temporal conditional probabilities scales the calculated posterior for all fault nodes
- Robust method for fault diagnosis given the fault-symptom causal relationships are defined correctly

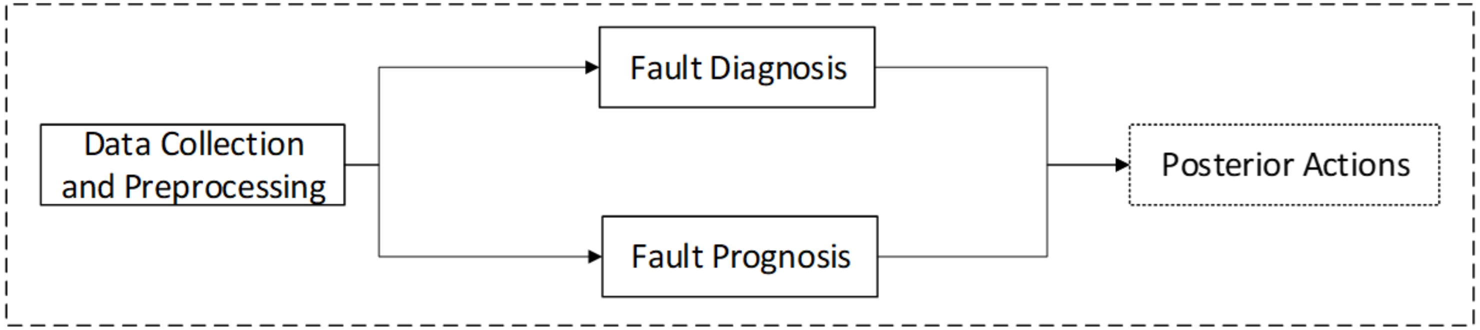
Task 3.1 Integration with Diagnosis

a) Fault diagnosis and prognosis conducted in series

b) Fault diagnosis and prognosis conducted in parallel



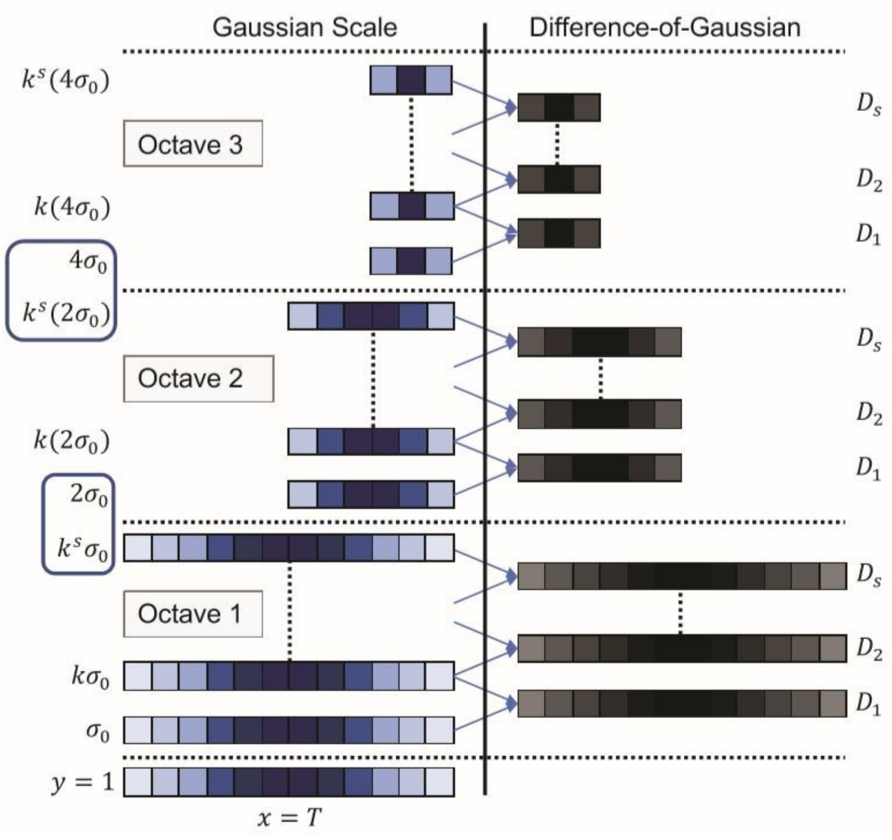
(a)



(b)

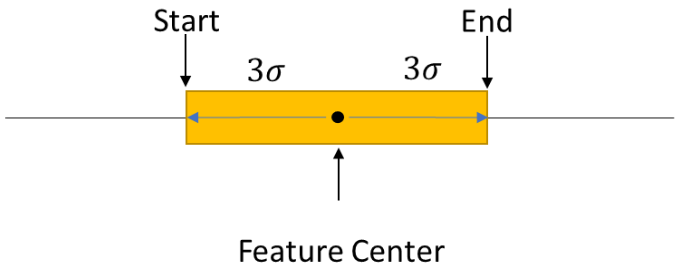
Task 3.2 RMT Variate Selection

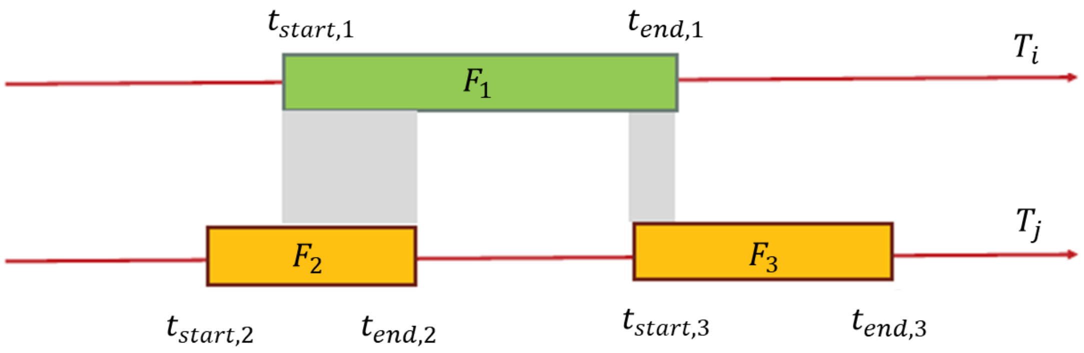
Generating DoG for single timeseries with length T



Garg and Candan, 2020

Temporal Scope Length





$$overlap(F_1, F_2) = min(t_{end,1}, t_{end,2}) - max(t_{start,1}, t_{start,2})$$

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Task 3.2 RMT Variate Selection

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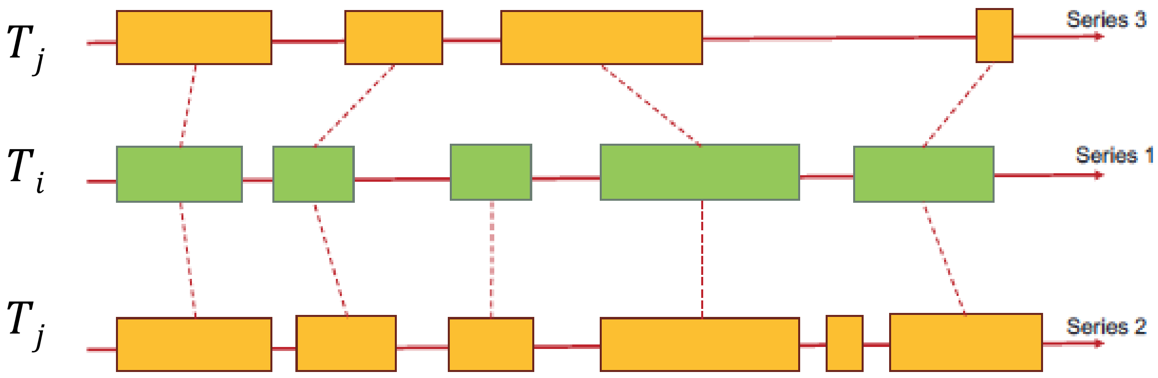
Variate Alignment

Given two variates, T_i and T_j with feature sets, $F_i = \{F_{i,1}, F_{i,2}, \ldots, |F_i|\}$ and $F_j = \{F_{j,1}, F_{j,2}, \ldots, |F_j|\}$, respectively, the temporal alignment of variate T_i against variate T_j is defined as:

$$TA(T_i|T_j) = \frac{\sum_{m=1}^{|F_i|} max_{n\epsilon(1,\ldots,|F_j|)}\, IA(\mathcal{F}_{i,m}, \mathcal{F}_{j,n})}{|F_i|}$$

Then define the variate alignment between two variates as:

$$VA(T_i|T_j) = TA(T_i|T_j) + TA(T_j|T_i)$$



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